

Large Language Models and Foundation Models for Time Series: A Survey and Outlook

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Abstract—Time series analysis underpins mission-critical decision systems across modern society, including industrial Internet of Things (IoT), computational finance, energy grid stabilization, meteorological forecasting, and clinical healthcare. Over the past five years, the field has undergone a profound paradigm shift from narrow, task-specific deep neural networks (such as specialized LSTMs and convolutional architectures) toward scalable foundation models. This evolution is bifurcated into two dominant architectural philosophies: (1) Native Time Series Foundation Models (TSFMs), which are trained ab initio on massive cross-domain time-series corpora comprising tens to hundreds of billions of observations (e.g., Chronos, TimesFM, MOIRAI, MOMENT, Lag-Llama, TTM, Timer, Sundial, Time-MoE, and Toto); and (2) Repurposed Large Language Models (LLM4TS), which repurpose frozen or parameter-efficiently adapted language transformers to temporal sequences through cross-modal token reprogramming, decimal prompting, or multimodal instruction tuning (e.g., Time-LLM, GPT4TS/OFA, LLMTime, TEMPO, and S2IP-LLM). In this survey, conducted strictly under the PRISMA 2020 systematic review protocol, we provide the first unified, comprehensive taxonomy and comparative synthesis spanning 2021 through 2026. We formalize the mathematical preliminaries of temporal representation, deconstruct core tokenization mechanics (patching vs. scalar quantization vs. lag vectors), analyze probabilistic formulation heads, examine empirical scaling behaviors and zero-shot transfer capabilities, and critically evaluate the central debate: Are billion-parameter language backbones truly necessary, or do compact specialized foundation models provide superior accuracy and operational efficiency? Finally, we conduct an audit of standardized benchmarks (e.g., GIFT-Eval), uncover evaluation pitfalls such as benchmark contamination and temporal leakage, and delineate open research directions toward multi-modal temporal intelligence.

Index Terms—Time series analysis, foundation models, large language models, deep learning, zero-shot forecasting, tokenization, scaling laws, PRISMA systematic review.



1 Introduction

Temporal data, systematically recorded as univariate or multivariate time series, serves as the fundamental perceptual substrate for understanding dynamic physical, economic, and biological systems. Across modern industries, time series forecasting, anomaly detection, imputation, and classification drive mission-critical workflows: stabilizing renewable energy power grids, optimizing multi-echelon retail supply chains, executing high-frequency financial risk arbitrage, predicting clinical sepsis in intensive care units, and monitoring aerospace telemetry [1].

Historically, time series modeling evolved through three distinct algorithmic epochs:

- 1) Statistical and Econometric Formulation (1970s–2010s): Dominated by classical stochastic processes, such as Auto-Regressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), Vector Autoregression (VAR), and Gaussian Processes. While mathematically rigorous and interpretable, these methods rely on linear assumptions and fail to capture complex nonlinear dependencies across cross-domain series.
- 2) Task-Specific Deep Neural Networks (2015–2022): Catalyzed by recurrent architectures (LSTMs, GRUs), temporal convolutional networks (TCNs), and specialized

Transformer variants (Informer, Autoformer, FEDformer, and PatchTST [2]). Although these architectures captured long-range nonlinear temporal correlations, each model was trained in isolation on a single, isolated dataset (e.g., ETT, Weather, Electricity), exhibiting virtually zero out-of-distribution generalization.

- 3) The Foundation Model Epoch (2022–Present): Propelled by the extraordinary success of pre-trained foundation models in Natural Language Processing (e.g., GPT-4, Llama) and Computer Vision, the temporal machine learning community has initiated a paradigm shift toward universal, cross-domain foundation models [3]–[5]. Rather than training narrow models from scratch for every new dataset, practitioners now seek models capable of strong zero-shot and few-shot transfer across unseen domains, frequencies, and channels.

1.1 Unique Challenges of Temporal Signals

Transferring the foundation model paradigm to time series is non-trivial due to stark physical and mathematical differences between temporal signals and natural language:

- **Absence of a Discrete Natural Vocabulary:** Unlike natural text, which is composed of finite, discrete lexical tokens with stable semantic meanings, continuous time series $\mathbf{x} \in \mathbb{R}^T$ consist of real-valued scalar measurements bounded only by sensor physics.
- **Extreme Heterogeneity in Sampling Rates and Dynamics:** Real-world time series exhibit widely divergent temporal

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scales—ranging from milliseconds in financial order books, to hourly power consumption, to daily retail demand, to quarterly economic indicators. A model must reconcile varying seasonalities without destructive aliasing.

- **Variable Dimensionality and Missing Values:** Time series data dynamically switch between univariate streams and arbitrary multivariate configurations ($C \geq 1$), frequently accompanied by irregular observation intervals, sensor failure dropouts, and structural trend shifts.

1.2 The Bifurcated Landscape: Native TSFMs vs. LLM4TS

To address these challenges, the research community has split into two competing architectural paradigms:

- 1) **Native Time Series Foundation Models (TSFMs):** Architectures designed and trained from scratch on massive multi-domain temporal corpora comprising tens to hundreds of billions of observations. Prominent exemplars include Amazon Chronos [3], [8], Google TimesFM [4], [9], Salesforce MOIRAI [5], [10], [11], CMU MOMENT [12], IBM TTM [13], Morgan Stanley/Mila Lag-Llama [14], and the THUML Timer family [15]–[18].
- 2) **Repurposed Large Language Models (LLM4TS):** Approaches that leverage the pre-trained weights, attention layers, and sequence modeling capabilities of multi-billion parameter language models (such as Llama, GPT-2, or GPT-4). Techniques include cross-modal token reprogramming (e.g., Time-LLM [19], GPT4TS/OFA [20], TEMPO [21]), direct numeric prompting with ASCII decimal strings (e.g., LLMTime [6]), and multimodal question-answering agents (e.g., Time-MQA [22], TimeOmni-1 [23]).

1.3 Key Contributions of This Survey

While several initial surveys have examined aspects of deep learning for time series, the rapid surge of foundation models from 2024 to 2026 demands a rigorous, systematic, and up-to-date synthesis. As highlighted in Table 1, this work delivers the following contributions:

- 1) **PRISMA 2020 Systematic Review Protocol:** We execute a strictly reproducible review protocol across scholarly APIs, logging 83 initial records, screening against standardized criteria, and synthesizing 41 core benchmark and foundation models.
- 2) **Rigorous Unified Taxonomy:** We establish an orthogonal five-dimensional taxonomy deconstructing models along: (i) architectural backbone, (ii) temporal tokenization, (iii) prediction head and uncertainty formulation, (iv) adaptation mechanics, and (v) operational task scope.
- 3) **Deep Comparative Synthesis of Native TSFMs vs. LLM4TS:** We analyze the theoretical and empirical tensions between compact native models (1M–700M parameters) and multi-billion parameter language adaptations (7B–70B parameters), examining computational efficiency, inference latency, and inductive bias alignment.
- 4) **Benchmark Audit, Data Contamination, and Scaling Laws:** We survey modern evaluation suites (e.g., GIFT-Eval [24]), expose systemic data leakage risks [25], analyze empirical power-law scaling in temporal models, and assess probabilistic calibration reliability [26].

- 5) **Open Research Agenda:** We articulate key open challenges, including irregular sampling equivariance, continuous flow matching, multimodal sensor fusion, and edge deployment.

1.4 Structure of the Survey

The remainder of this survey is organized as follows: Section 2 formalizes mathematical preliminaries and foundational objectives. Section 3 presents our comprehensive taxonomy and comparative design table. Section 4 deconstructs Native TSFMs. Section 5 investigates Repurposed LLM methods. Section 6 evaluates benchmark suites, scaling laws, and critical debates. Section 7 delineates open challenges, and Section 8 concludes the paper.

2 Background and Preliminaries

In this section, we establish the formal mathematical definitions, notation, and objective functions that govern modern temporal foundation models and large language model adaptations.

2.1 Formal Problem Definitions

2.1.1 Temporal Sequences

A univariate time series of length T is defined as an ordered sequence of real-valued scalar observations:

$$\mathbf{x} = [x_1, x_2, \dots, x_T]^\top \in \mathbb{R}^T, \quad (1)$$

where $x_t \in \mathbb{R}$ represents the measurement recorded at discrete timestamp $t \in \{1, \dots, T\}$.

A multivariate time series comprises C co-evolving channels or variates observed synchronously across T timestamps:

$$\mathbf{X} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}]^\top \in \mathbb{R}^{C \times T}, \quad (2)$$

where $\mathbf{X}_{:,t} \in \mathbb{R}^C$ is the vector of all variate measurements at timestamp t .

2.1.2 Channel Modeling Paradigms

When processing multivariate series $\mathbf{X}$, architectures adopt one of two primary structural paradigms:

- **Channel-Independence (CI):** Introduced prominently in PatchTST [2] and Timer [15], CI decomposes the multivariate series into C distinct univariate sequences:

$$\mathbf{X} \mapsto \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}\}, \quad (3)$$

which are fed into the shared model backbone independently with tied parameters. CI eliminates the risk of overfitting spurious cross-channel correlations and natively handles variable channel counts C at inference.

- **Channel-Mixing (CM):** Models such as MOIRAI [5] and TTM [13] employ full cross-variate attention or specialized channel-mixing MLPs:

$$\mathbf{H} = \text{Attn}_{\text{channel}}(\mathbf{X}) \in \mathbb{R}^{C \times T \times D}, \quad (4)$$

explicitly capturing physical inter-sensor dependencies at the cost of higher computational complexity.

2.2 Tokenization and Input Representation

Because continuous temporal values lack an inherent lexicon, models deploy specialized mapping operators $\mathcal{T} : \mathbb{R}^* \rightarrow \mathbb{R}^D$ to construct transformer-compatible token embeddings.

TABLE 1
Comparison of This Survey with Related Surveys on Large Temporal Models.

Survey Reference	Year	PRISMA	Native TSFMs	Repurposed LLMs	Critiques & Audits	Primary Focus & Distinguishing Scope
Wen et al. [1]	2023	✗	✗	✗	✗	Survey of specialized Transformer architectures on single datasets.
Jin et al. [1]	2024	✗	Partial	✓	Partial	Broad survey covering both general time series and spatio-temporal graphs.
Gruver et al. [6]	2023	✗	✗	✓	✗	Empirical investigation of prompting LLMs as zero-shot forecasters.
Fons et al. [7]	2024	✗	✗	✓	✓	Empirical critique testing whether language backbones add genuine inductive value.
This Survey	2026	✓	✓	✓	✓	Comprehensive PRISMA systematic review (2021–2026); unifying Native TSFMs, Repurposed LLMs, tokenization trade-offs, scaling laws, and leakage audits.

2.2.1 Subseries Patching

Rather than embedding individual scalar time-steps (which leads to quadratic attention overhead $\mathcal{O}(T^2)$ and severe information sparsity), the majority of modern TSFMs (e.g., TimesFM [4], Timer [15], MOMENT [12]) partition a sequence $\mathbf{x}$ into N contiguous subseries patches of length P with stride S :

$$N = \left\lfloor \frac{T-P}{S} \right\rfloor + 1. \quad (5)$$

The i -th patch $\mathbf{p}_i = [x_{(i-1)S+1}, \dots, x_{(i-1)S+P}]^\top \in \mathbb{R}^P$ is mapped to a D -dimensional latent vector $\mathbf{e}_i \in \mathbb{R}^D$ via an affine projection:

$$\mathbf{e}_i = \mathbf{p}_i \mathbf{W}_p + \mathbf{b}_p + \mathbf{e}_{\text{pos},i}, \quad (6)$$

where $\mathbf{W}_p \in \mathbb{R}^{P \times D}$, $\mathbf{b}_p \in \mathbb{R}^D$, and $\mathbf{e}_{\text{pos},i} \in \mathbb{R}^D$ denotes a learnable or sinusoidal positional encoding. Patching preserves local temporal shape dynamics and compresses sequence length from T to $N \ll T$.

2.2.2 Scalar Quantization and Categorical Binning

In contrast to continuous linear patching, Amazon Chronos [3] introduces a pure language-analog tokenization. Given an input sequence $\mathbf{x}$, it computes instance-level scale statistics (μ, σ) via mean scaling:

$$\tilde{x}_t = \frac{x_t - \mu}{\sigma + \epsilon}, \quad \mu = \frac{1}{T} \sum_{t=1}^T x_t, \quad \sigma = \frac{1}{T} \sum_{t=1}^T |x_t - \mu|. \quad (7)$$

The normalized scalars $\tilde{x}_t$ are discretized into K uniform bins over $[-B, B]$:

$$q_t = \text{clip} \left(\left\lfloor \frac{\tilde{x}_t - (-B)}{2B/K} \right\rfloor, 0, K-1 \right), \quad (8)$$

where $q_t \in \{0, 1, \dots, K-1\}$ acts as a discrete categorical token id, enabling standard cross-entropy language modeling.

2.3 Downstream Task Formulations

2.3.1 Zero-Shot and Supervised Forecasting

Given an historical context window $\mathbf{X}_{1:T} \in \mathbb{R}^{C \times T}$, the forecasting objective is to predict the future sequence of length H :

$$\hat{\mathbf{Y}} = \hat{\mathbf{X}}_{T+1:T+H} \in \mathbb{R}^{C \times H}. \quad (9)$$

In probabilistic forecasting, the model estimates the conditional predictive distribution:

$$p(\mathbf{Y} \mid \mathbf{X}_{1:T}) = \prod_{h=1}^H p(y_{T+h} \mid \mathbf{X}_{1:T}, y_{T+1:T+h-1}). \quad (10)$$

This distribution is parameterized either parametrically (e.g., Student's t parameters in Lag-Llama [14] or Gaussian mixtures in MOIRAI [5]) or non-parametrically through categorical bin logits (Chronos [3]).

2.4 Pre-training Objectives

2.4.1 Autoregressive Generation (Next-Patch / Next-Token)

Decoder-only foundation models (TimesFM [4], Timer [15], Chronos [3]) optimize an autoregressive likelihood:

$$\mathcal{L}_{\text{AR}}(\Theta) = - \sum_{i=1}^N \log p_{\Theta}(\mathbf{p}_{i+1} \mid \mathbf{p}_1, \dots, \mathbf{p}_i). \quad (11)$$

For continuous regression heads (Timer, TimesFM), this is computed using Mean Squared Error (MSE) or Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{reg}}(\Theta) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_{i+1} - \hat{\mathbf{p}}_{i+1}\|_2^2. \quad (12)$$

2.4.2 Masked Autoencoding (MAE)

Encoder-based foundation models (MOMENT [12], PatchTST [2]) apply a binary masking vector $\mathbf{m} \in \{0, 1\}^N$ over patches, optimizing patch reconstruction over the masked set $\mathcal{M} = \{i \mid m_i = 1\}$:

$$\mathcal{L}_{\text{Mask}}(\Theta) = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\mathbf{p}_i - f_{\Theta}(\mathbf{X}_{\setminus \mathcal{M}})\|_2^2. \quad (13)$$

Masked autoencoding yields bidirectional temporal representations particularly suited for downstream classification, anomaly detection, and imputation.

3 Taxonomy and Landscape Overview

To organize the rapid proliferation of temporal foundation models and large language model adaptations, we present an orthogonal multi-dimensional taxonomy. As illustrated in Figure 1, the landscape is structured along five fundamental axes: (1) Core Paradigm, (2) Architectural Backbone, (3) Temporal Tokenization, (4) Prediction Head & Uncertainty, and (5) Operational Scope & Tasks.

3.1 Taxonomy Dimensions

3.1.1 Dimension 1: Core Paradigm

- **Native Time Series Foundation Models (Native TSFMs):** Models built from scratch and trained directly on large-scale collections of empirical or synthetic time-series data. Weights and latent dimensions are optimized specifically for temporal signals without textual preconceptions (e.g., Chronos [3], TimesFM [4], MOIRAI [5], ForecastPFN [27], TiRex [28], FLAME [29], t_0 [30]).
- **Repurposed Large Language Models (LLM4TS):** Models that adopt pre-existing weights from language or vision transformers (e.g., Llama-2/3, GPT-2, CLIP), bridging continuous temporal modalities into text or visual embedding space via reprogramming layers, prompt prefixes, or cross-attention adapters (e.g., Time-LLM [19], GPT4TS [20], TEST [31], CALF [32], UniTime [33], VisionTS [34], Time-VLM [35]).
- **Evaluation Frameworks and Benchmark Audits:** Standardized benchmarking suites and diagnostic harnesses that audit generalization, cross-domain transfer, and benchmark contamination (e.g., GIFT-Eval [24], SciTS [36], Insight Miner [37], Time-MMD [38]).

3.1.2 Dimension 2: Architectural Backbone

- **Decoder-Only Autoregressive Transformers:** Optimized for sequential next-token or next-patch generation (e.g., TimesFM [4], Timer [15], Chronos [3], Sundial [17], t_0 [30]).
- **Masked Encoder Transformers:** Optimized for bidirectional temporal representations via masked autoencoding (e.g., MOMENT [12], PatchTST [2], VisionTS [34]).
- **Encoder-Decoder Transformers:** Decoupling context representation from horizon generation (e.g., MOIRAI [5], TTM [13], UniTS [39]).
- **Sparse Mixture-of-Experts (MoE):** Dynamically routing tokens to specialized feed-forward experts to scale model capacity while maintaining constant inference FLOPs (e.g., Time-MoE [40], Moirai-MoE [10], Timer-S1 [18]).
- **Continuous Flow and Legendre Memory Networks:** Employing continuous neural ODEs or orthogonal polynomial

bases to preserve multi-scale memory (e.g., FLAME [29], FlowState [41]).

3.1.3 Dimension 3: Temporal Tokenization

- **Uniform Subseries Patching:** Continuous linear projection of non-overlapping or strided time-step windows into dense embedding space (e.g., PatchTST [2], Timer [15]).
- **Scalar Discretization / Binning:** Quantizing scaled continuous observations into finite categorical tokens, allowing standard NLP cross-entropy training (Chronos [3]).
- **Decimal String Tokenization:** Casting continuous numbers into ASCII strings (e.g., "12.4"), processed directly by off-the-shelf text tokenizers (PromptCast [42], LLMTime [6]).
- **Multi-Resolution and Multiscale Mixing:** Dynamically decomposing and mixing patches across frequencies to resolve multi-scale seasonalities (TimeMixer [43], TimeMixer++ [44], TTM [13]).
- **Visual Rendering Tokenization:** Rendering 1D numerical trajectories into 2D pixel line plots for visual encoders (VisionTS [34], Time-VLM [35]).

3.1.4 Dimension 4: Prediction Head and Uncertainty

- **Categorical Probability Distribution:** Non-parametric probability vectors over quantized bins (Chronos [3]).
- **Parametric Mixture Heads:** Estimating parameters (μ, σ, ν) of explicit probability densities, such as Student's t or Gaussian mixtures (Lag-Llama [14], MOIRAI [5]).
- **Deterministic Mean Squared Regression:** Direct linear or MLP projection minimizing mean squared or absolute error.
- **Continuous Flow Matching and Diffusion:** Formulating generation as an equivariant continuous-time velocity field (FlowState [41], FLAME [29]).
- **Quantile Regression Heads:** Directly outputting discrete quantile trajectories to provide non-parametric prediction intervals (t_0 [30], TimesFM [4], Toto 2.0 [45]).

4 Native Time Series Foundation Models

Native Time Series Foundation Models (TSFMs) represent the purist approach to temporal foundation modeling: architectures designed from first principles and trained directly on massive corpora of continuous time-series data without linguistic baggage. In this section, we examine the primary architectural branches, data curation pipelines, and empirical scaling mechanisms.

4.1 Autoregressive Decoder-Only Architectures

Autoregressive models frame time-series modeling as sequential generation, predicting subsequent time-steps or patches conditioned on historical context.

4.1.1 The Chronos Family: Discretization as Linguistic Analogy

Amazon's Chronos [3] represents one of the most radical departures from conventional continuous time-series modeling. Rather than predicting real numbers via regression losses, Chronos asks: Can time series be modeled identically to language tokens? By applying instance-level mean scaling followed by uniform quantization into $K = 4096$ bins, continuous values are transformed into integer token indices.

Taxonomy of Foundation Models and Large Language Models for Time Series

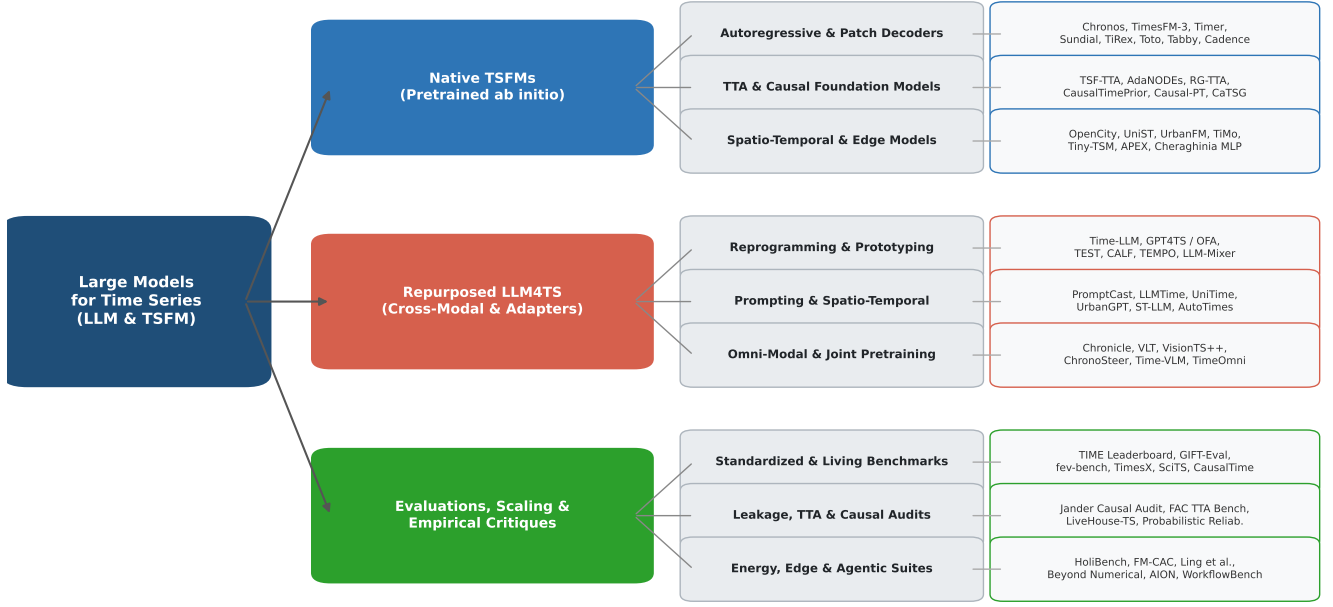


Fig. 1. Comprehensive taxonomy of foundation models and large language models for time series, illustrating core paradigms, architectural backbones, tokenization mechanics, and representative models.

TABLE 2
Comprehensive Comparison of Representative Time Series Foundation Models and LLM Methods.

Model	Group	Venue/Yr	Paradigm	Backbone	Tokenization	Params	Pretraining Corpus	Uncertainty	Code / Status
Chronos [3]	Amazon	TMLR'24	Native TSFM	Decoder	Quantized Bins	20M–710M	TSMix (84B obs)	Categorical	amazon/chronos
TimesFM [4]	Google	ICML'24	Native TSFM	Decoder	Patch ($P = 32$)	200M	Real+Synth (100B)	Quantile	google/timesfm
MOIRAI [5]	Salesforce	ICML'24	Native TSFM	Enc-Dec	Multi-patch (8–128)	14M–311M	LOTSa (27B obs)	Mixture Param	salesforce/uni2ts
MOMENT [12]	CMU	ICML'24	Native TSFM	Encoder	Patch ($P = 512$)	385M	TS-Pile (1B obs)	Recon Head	cmu/moment
Lag-Llama [14]	Mila / MS	ICML'24	Native TSFM	Decoder	Lag Vectors	2.4M	Monash (7.9K ser)	Student's t	mila/lag-llama
TTM [13]	IBM	NeurIPS'24	Native TSFM	Enc-Dec	Adaptive Patch	1M–8M	Multi-domain Pub	Deterministic	ibm/granite-tsfm
Timer [15]	THUML	ICML'24	Native TSFM	Decoder	Subseries Patch	84M	UTSD (1B pts)	Next-Patch	thuml/Timer
Timer-XL [16]	THUML	NeurIPS'24	Native TSFM	Decoder	Long-Ctx Patch	84M	UTSD-2	Next-Patch	thuml/Timer-XL
Sundial [17]	THUML	arXiv'25	Native TSFM	Decoder	Multi-rate Patch	1.5B	UTSD-3 (>10B)	Continuous	thuml/Sundial
Time-MoE [40]	Ming Jin	ICLR'25	Native TSFM	MoE	Multi-res Patch	2.4B	Time-300B (300B)	MoE Contin	Time-MoE
Timer-S1 [18]	THUML	arXiv'26	Native TSFM	MoE	Serial Patch	8.3B	UTSD-3	MoE Regress	thuml/Timer-S1
ForecastPFN [27]	Abacus/CMU	NeurIPS'23	Native TSFM	Transformer	In-Context PFN	5M	Synthetic Prior	Quantile	abacus/ForecastPFN
t_0 [30]	Invenia/ETH	arXiv'26	Native TSFM	Axial Attn	Context Patch	102M–256M	Pub+Synth	Quantile	Proprietary
Toto 2.0 [45]	Datadog	arXiv'26	Native TSFM	Decoder	Wavelet Tokens	1.2B	Telemetry (1.5T)	Quantile	Proprietary
Time-LLM [19]	Ming Jin	ICLR'24	LLM4TS	Llama-7B	Reprogrammed	7B	Pretrained LLM	Deterministic	KimMeen/Time-LLM
GPT4TS [20]	DAMO	NeurIPS'23	LLM4TS	GPT-2	Cross-Modal	Unreported	Pretrained GPT-2	Deterministic	DAMO/GPT4TS
TEST [31]	PKU/Alibaba	NeurIPS'24	LLM4TS	Llama/BERT	Text Prototype	7B	Pretrained LLM	Deterministic	Proprietary
CALF [32]	THU/DAMO	KDD'24	LLM4TS	Llama-7B	Distilled Patch	7B	Pretrained LLM	Deterministic	Hank0626/CALF
LLMTime [6]	NYU	NeurIPS'23	LLM4TS	GPT-4/Llama	Decimal ASCII	70B+	Pretrained LLM	Sampling Logit	ngruver/llmtime
TEMPO [21]	Ming Jin	ICLR'24	LLM4TS	GPT-2	Decomposed	Unreported	Pretrained GPT-2	Decomposed	caocao/TEMPO
AutoTimes [46]	THUML	NeurIPS'24	LLM4TS	Llama-2-7B	Point-wise	7B	Pretrained LLM	Autoregress	thuml/AutoTimes
Time-MQA [22]	Ming Jin	ACL'25	LLM4TS	Llama-3-8B	Patch + Prompt	8B	Multi-domain MQA	Text/Value	Proprietary

Chronos trains standard T5 architectures (ranging from 20M to 710M parameters) solely with the cross-entropy objective. The predictive distribution over future horizons is obtained by autoregressively sampling categorical bin logits and dequantizing them back into real space. This approach has three key advantages:

- 1) It sidesteps parametric distribution assumptions (e.g., Gaussianity), naturally expressing multi-modal, skewed, or heavy-tailed predictive distributions.
- 2) It enables direct reuse of optimized language transformer implementations (such as FlashAttention and KV-caching).
- 3) In Chronos-2 [8], the architecture is extended to multi-

variate series via grouped channel tokenization and hybrid continuous residual heads.

4.1.2 Google TimesFM: Multi-Patch Output Generation

Unlike Chronos's scalar quantization, Google's TimesFM [4] operates directly on continuous numerical values using continuous patching. TimesFM employs an input patch size of $P = 32$ and an output patch size of $H = 128$. By predicting an entire multi-step block in a single forward step, TimesFM drastically reduces autoregressive unrolling steps, curbing error accumulation and inference latency. TimesFM is trained on a massive corpus of 100B time-series data points collected from Google Trends, Wikipedia pageviews, and synthetic

ARMA processes. Recent extensions examine in-context fine-tuning [9], demonstrating that supplying short conditioning examples within the context window improves zero-shot adaptation. Subsequently, TimesFM-3 scales the architecture to 330M parameters pre-trained over 1T time points, introducing native multivariate forecasting via stacked variate attention and iterative reversible instance normalization (RevIN). Demonstrating the industrial viability of large-scale foundation forecasters, Tacconelli [47] introduces Cadence, pairing TimesFM-3 with an adaptive arithmetic coder to execute error-bounded lossy compression on enterprise telemetry, guaranteeing user-defined maximum deviation bounds while achieving state-of-the-art bit-rate reductions.

4.1.3 The THUML Timer Family and Output Scaling

The THUML research group (led by Mingsheng Long at Tsinghua University) has systematically advanced generative time series pretraining:

- Timer [15]: Conceptualizes a single-series sequential next-patch regression paradigm. Timer is pretrained on the Unified Time Series Dataset (UTSD), comprising over 1B observations across diverse domains. Timer establishes that channel-independent single-series formatting enables universal transfer across arbitrary multivariate configurations.
- Timer-XL [16]: Introduces long-context transformer modeling for temporal signals, scaling context lengths from standard 512 patches to extreme contexts, enabling the model to capture multi-year periodicities without context truncation.
- Sundial [17]: Scales the model to 1.5B parameters pretrained on UTSD-3 (> 10B observations), introducing multi-rate adaptive patching to handle heterogeneous sampling frequencies seamlessly.
- Timer-S1 [18]: Reaches 8.3B parameters via sparse Mixture-of-Experts routing, demonstrating the viability of serial parameter scaling in native temporal models.
- YingLong [48]: Explores test-time output scaling in pre-trained forecasting models by introducing delayed chain-of-thought (CoT) tokens. Rather than predicting future values in a single forward pass, YingLong autoregressively generates intermediate trend and regime change tokens before decoding target values, demonstrating that computational output scaling laws apply to temporal reasoning.

4.1.4 Context-Conditioned and Dual-Horizon Autoregressive Models

Standard autoregressive models often struggle with the structural tension between resolving high-frequency short-term fluctuations and maintaining long-range macro trajectories. Recent 2025–2026 architectures address this via explicit context-conditioning:

- TiRex [28]: Auer et al. (JKU Linz / ELLIS) propose an asymmetric dual-horizon in-context architecture. TiRex isolates local fine-grained patches from global horizon summaries, using memory compaction to preserve multi-scale context without quadratic computational blowup.
- t_0 Family [30]: Meyer et al. introduce t_0 , releasing open-weights checkpoints t_0 -alpha (102M) and t_0 -beta (256M). Departing from channel-independent univariate approaches, t_0 explicitly conditions predictions on historical

targets, past dynamic covariates, and known-future covariates (e.g., promotional calendars, weather forecasts). By alternating axial self-attention along the temporal axis and cross-variate attention across feature channels, t_0 attains aggregate CRPS of 0.4738 and MASE of 0.6865 on GIFT-Eval, standing within 4% of the top zero-shot forecaster.

- Toto Family [45], [49]: Datadog introduced Toto (151M parameters) [49], trained on 1T cloud infrastructure and application metrics to deliver joint zero-shot quantile forecasting and out-of-distribution anomaly scoring. Toto 2.0 [45] subsequently scales from 4M to 2.5B parameters using Maximal Update Parametrization (μ P), achieving state-of-the-art results on GIFT-Eval and the contamination-resistant TIME benchmark.
- Tabby Open Recipe [50]: Addressing the pervasive opacity of proprietary pretraining mixtures, Xie et al. introduce Tabby (120M parameters), releasing an open-source, long-context probabilistic foundation model accompanied by a fully reproducible data curation pipeline and an open 150B-token pretraining corpus.

4.1.5 Lag-Llama: Probabilistic Modeling with Lag Embeddings

Developed by Morgan Stanley and Mila, Lag-Llama [14] adapts the Llama decoder backbone by substituting word embeddings with lag feature vectors. Specifically, it computes lags corresponding to standard calendar frequencies (e.g., quarter-hour, day, week, year). Lag-Llama outputs the parameters of a Student’s t distribution, achieving robust probabilistic calibration on out-of-distribution Monash datasets.

4.2 Prior-Data Fitted Networks (PFNs) and In-Context Meta-Learning

A radically distinct paradigm in native foundation modeling is the Prior-Data Fitted Network (PFN) and in-context learning framework, which foregoes empirical training corpora or iterative gradient steps entirely in favor of synthetically generated structural priors:

- ForecastPFN [27]: Dooley et al. demonstrate that a transformer trained exclusively on synthetic time series generated from a Bayesian mixture of Gaussian processes, ARMA models, and dynamical equations can approximate posterior predictive distributions in a single forward pass. ForecastPFN requires zero gradient steps or hyperparameter tuning at inference time, achieving competitive zero-shot accuracy across diverse real-world datasets.
- TabPFN-TS [51]: Extending tabular foundation models (TabPFN-v2) to the temporal domain, Müller et al. propose TabPFN-TS (11M parameters). By encoding time series as tabular regression tasks conditioned on causal lag features and exogenous metadata, TabPFN-TS achieves state-of-the-art accuracy on covariate-informed forecasting across GIFT-Eval and fev-bench, proving that in-context synthetic priors offer extreme parameter efficiency.
- In-Context Time Series Predictor (ICTSP) [52]: Lu et al. (ICLR 2025) formalize in-context regression for continuous temporal sequences. By structuring historical context and query sequences within an interleaved prompt-query attention mechanism, ICTSP (42M parameters) executes zero-gradient meta-adaptation across disparate sampling

frequencies and noise regimes without updating network weights.

4.3 Masked Encoders, Multiscale Mixers, and Multi-Task Architectures

4.3.1 Salesforce MOIRAI: Any-Variate Cross-Attention

Salesforce’s MOIRAI [5] addresses two central limitations of previous models: rigid patch sizes and channel-independence. MOIRAI introduces Any-Variate Attention, flattening an arbitrary multivariate sequence $\mathbf{X} \in \mathbb{R}^{C \times T}$ into a sequence of variate-tagged patch tokens. Furthermore, it employs a multi-patch projection head capable of simultaneously processing patch lengths of 8, 16, 32, 64, and 128. MOIRAI is trained on the Large-scale Open Time Series Archive (LOTSAs), containing 27B observations across 9 distinct domains. The model outputs mixtures of parametric distributions, achieving top-tier ranking on zero-shot benchmarks. Moirai 2.0 [11] and Moirai-MoE [10] further extend this framework to sparse routing with 1.1B parameters.

4.3.2 CMU MOMENT and UniTS: Universal Multi-Task Representation

While most TSFMs focus exclusively on forecasting, CMU’s MOMENT [12] investigates foundation models for universal multi-task analysis (forecasting, classification, anomaly detection, and imputation). Based on a masked T5 encoder (385M parameters), MOMENT is pretrained on the Time-series Pile (1B observations) using masked patch reconstruction. Similarly, Harvard’s UniTS [39] unifies multi-task modeling through a single transformer backbone. By combining shared attention operators with task-specific prompt vectors and dynamic gating networks, UniTS demonstrates that cross-task representations mutually reinforce classification and forecasting generalization.

4.3.3 Lightweight Mixers and Exogenous Variable Modeling

Challenging the premise that foundation models require massive parameter scales:

- Tiny Time Mixers (TTM) [13]: IBM Research introduced TTM (1M–8M parameters), utilizing lightweight multi-level TSMixer blocks with adaptive resolution sampling. TTM out-executes models orders of magnitude larger while running inference in milliseconds on standard edge CPUs.
- TimeMixer and TimeMixer++ [43], [44]: Wang et al. formulate multiscale mixing architectures. TimeMixer decomposes signals into downsampled representations to disentangle multiscale seasonalities, while TimeMixer++ extends this principle into a general temporal pattern machine supporting cross-domain transfer.
- TimeXer [53]: Specifically targets exogenous variables by learning cross-attention maps between target end-points and exogenous features, avoiding noise propagation from uninformative covariates.
- Kairos [54]: Introduces predictability-stratified pretraining (PreSTS), filtering chaotic or non-deterministic temporal noise during self-supervised training to ensure foundation models focus representational capacity on predictable temporal dynamics.
- LightGTS [55]: Wang et al. (ICML 2025) challenge the assumption that cross-domain generalizability requires heavy

parameter footprints. LightGTS (5.8M parameters) introduces dual-path hierarchical patching and adaptive parameter allocation, matching the zero-shot accuracy of foundation models two orders of magnitude larger while running inference directly on resource-constrained edge microcontrollers.

4.4 Continuous Flow Matching and Polynomial Memory Networks

A vital 2025 frontier replaces discrete tokenization with continuous-time dynamical systems:

- FLAME [29]: Wu et al. introduce Flow-Enhanced Legendre Memory Models. FLAME compresses long continuous temporal histories onto orthogonal Legendre polynomial bases, avoiding memory degradation over extended sequences, while modeling predictive distributions via continuous normalizing flows.
- FlowState [41]: King et al. formulate time series forecasting via continuous-time ODEs and flow matching. FlowState achieves strict sampling-rate equivariance, allowing zero-shot inference across irregularly sampled or arbitrary continuous time-steps without interpolation artifacts. Despite its compact footprint (12M parameters), FlowState achieves top-tier CRPS on GIFT-Eval.
- EIDOS [56]: Zhou et al. (Ming Jin’s group, 2026) introduce latent-space predictive learning for temporal foundation models via a Joint-Embedding Predictive Architecture (JEPA). Rather than optimizing token-level mean-squared reconstruction—which forces the model to expend capacity memorizing stochastic high-frequency jitter—EIDOS predicts future dynamics within a learned latent metric space, yielding semantically structured representations that transfer robustly across unseen domains.

4.5 Mixture-of-Experts (MoE) and Scaling to Billions

A major architectural breakthrough between 2024 and 2026 is the deployment of Sparse Mixture-of-Experts (MoE) in time series. In Time-MoE [40] (Ming Jin’s group), the model scales to 2.4B parameters while only activating a subset of experts per patch token. Trained on Time-300B (300B time points), Time-MoE demonstrates clear empirical power-law scaling in validation loss, proving that scaling laws discovered in NLP hold equally for native temporal representations when sufficient data volume is supplied. Similarly, Moirai-MoE [10] (1.1B) and Timer-S1 [18] (8.3B) substantiate that sparse MoE routing allows foundation models to specialize on domain-specific dynamics (e.g., high-frequency finance vs. slow-moving climate trends) while keeping inference latency within practical bounds.

4.6 Parameter-Efficient Fine-Tuning vs. In-Context Adaptation

A central dilemma in deploying foundation models is choosing between zero-shot inference, in-context learning, parameter-efficient fine-tuning (PEFT), and full fine-tuning. Table 3 categorizes the core adaptation paradigms across efficiency, computational overhead, and empirical generalization trade-offs.

Empirical synthesis reveals two critical principles governing downstream adaptation:

TABLE 3
Systematic Comparison of Downstream Adaptation Paradigms for Time Series Foundation Models.

Adaptation Paradigm	Exemplar Models	Trainable Params	Gradient Steps	Memory Overhead	Data Regime / Sample Efficiency	Generalization & Stability
Zero-Shot Transfer	Chronos-2 [8], TimesFM [4], t_0 [30], Toto 2.0 [45]	0%	0	Baseline (KV-Cache only)	Zero target samples required; immediate out-of-the-box deployment	Preserves broad cross-domain calibration; bounded by pretraining domain overlap
In-Context Meta-Learning	ForecastPFN [27], TabPFN-TS [51], ICTSP [52], TimesFM-IC [9]	0%	0	Low-Medium (Expanded context window)	Extreme few-shot (5-50 context sequences); zero fine-tuning FLOPs	Immune to catastrophic forgetting; limited by maximum attention context length
Parameter-Efficient (PEFT / LoRA / Prefix)	TTM [13], CALF [32], Time-LLM [19], OFA [20]	0.05%-2.0%	Few (50-500)	Low (< 10% activation memory)	Highly sample-efficient (5%-10% target dataset matches full training)	Mitigates representation collapse via low-rank constraints; retains core backbone weights
Full-Parameter Fine-Tuning	MOMENT [12], PatchTST [2], Timer [15]	100%	Many (1K-10K)	High (4× model size for optimizer states)	Requires substantial target history (100% dataset); prone to overfitting on small data	Achieves lowest in-domain MSE on large stationary datasets; suffers catastrophic forgetting

- 1) The In-Context Learning Frontier: Prior-Data Fitted Networks (TabPFN-TS, ForecastPFN) and meta-learning predictors (ICTSP) demonstrate that conditioning on prompt-target pairs inside the attention context matches 100-step gradient descent while executing in a single forward pass, providing an ideal mechanism for real-time edge streaming where backpropagation is computationally prohibited.
- 2) PEFT Prevents Semantic Collapse: As demonstrated in CALF [32] and TTM [13], tuning less than 1% of weights via low-rank matrices or adaptive resolution prefix blocks yields within 2% of full fine-tuning performance on target benchmarks while completely insulating the model against catastrophic representation drift.

4.7 Spatio-Temporal Foundation Models: Unifying 1D Streams and Graph Topologies

A pervasive frontier in foundation modeling is extending beyond isolated 1D univariate or channel-independent series toward structured spatio-temporal systems. In transportation networks, urban human mobility, power distribution grids, and planetary meteorological monitoring, continuous observation streams are inextricably situated upon complex underlying geometric topologies [57].

Formally, a spatio-temporal system is represented as a spatial topology $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes a set of $N = |\mathcal{V}|$ spatial sensor nodes or grid cells, $\mathcal{E}$ represents physical or learned relational edges, and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the spatial adjacency matrix (or spatial distance metric). Given historical observations $\mathbf{X}_{1:T} = [\mathbf{x}_1, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times N \times D_{\text{in}}}$, the spatio-temporal foundation objective requires forecasting future states $\mathbf{X}_{T+1:T+H} \in \mathbb{R}^{H \times N \times D_{\text{out}}}$ across both seen and entirely unseen spatial topologies $\mathcal{G}^*$:

$$P(\mathbf{X}_{T+1:T+H} \mid \mathbf{X}_{1:T}, \mathcal{G}) = \prod_{h=1}^H P(\mathbf{x}_{T+h} \mid \mathbf{X}_{1:T+h-1}, \mathbf{A}). \quad (14)$$

Traditional deep spatio-temporal architectures (e.g., STGCN, DCRNN) relied on fixed static spatial graph Laplacian operators and task-specific training, catastrophically failing when deployed zero-shot across new sensor networks or different urban geographies. Modern Spatio-Temporal Foundation Models (STFMs) circumvent this structural barrier through four primary paradigms:

- Dual Spatio-Temporal Self-Attention with Graph Biases (OpenCity): Li et al. [58] present OpenCity, an open-world

foundation model for urban traffic forecasting. OpenCity decomposes representations into decoupled spatial and temporal transformer layers. To eliminate reliance on rigid fixed adjacency matrices, OpenCity injects generalized graph wavelet positional encodings and relative spatial distance embeddings into multi-head self-attention. Pre-trained on multi-city traffic networks comprising over 50M observations, OpenCity demonstrates remarkable zero-shot transfer across previously unobserved metropolitan road networks.

- Prompt-Empowered Universal Masked Autoencoding (UniST): Yuan et al. [59] introduce UniST (Tsinghua FIB Lab), moving beyond scenario-specific architectures. UniST formulates spatio-temporal forecasting through unified generative masked pretraining. To accommodate heterogeneous sampling rates, variable sensor densities, and semantic domain shifts (e.g., traffic velocity vs. pedestrian flow vs. bike sharing), UniST injects knowledge-guided continuous prompt tokens into the transformer backbone, guiding the shared model to dynamically adapt spatial-temporal priors without altering backbone parameters.
- Spatio-Temporal Diffusion Transformers (UrbanDiT): Li et al. [60] establish UrbanDiT, scaling Diffusion Transformers (DiT) to open-world spatio-temporal systems. UrbanDiT tokenizes both irregular spatial graphs and regular spatial raster grids into unified spatio-temporal patches. By combining reverse-time score-based diffusion denoising with task prompt embeddings, UrbanDiT unifies diverse downstream tasks—including bidirectional forecasting, spatial imputation, and trajectory extrapolation—under a single generative score network.
- Minimalist Scaling and Inductive Regularization (UrbanFM): Chen et al. [61] present UrbanFM, scaling spatio-temporal foundation models up to 120M parameters. UrbanFM challenges over-engineered spatial graph heuristics, proving that minimalist self-attention with constrained inductive biases and MiniST tokenization (which normalizes variance across disparate city sensors) achieves superior scaling laws and cross-city zero-shot transfer compared to complex graph convolution-transformer hybrids.
- Satellite Image Time Series Foundation Models (TiMo): Qin et al. [62] introduce TiMo, scaling spatio-temporal foundation architectures to Satellite Image Time Series (SITS). Departing from standard vision transformers that process static frames independently, TiMo incorporates a

hierarchical spatio-temporal gyroscope attention mechanism that captures evolving multi-scale seasonal patterns across planetary terrain. Pre-trained on the MillionST dataset comprising one million satellite image phases across 100,000 geographic locations over five years, TiMo unifies deforestation monitoring, crop classification, and flood risk segmentation under a single masked autoencoding foundation.

- Unified Conceptual Vision and Generalization Challenges: Goodge et al. [57] formulate a comprehensive taxonomy for STFMs, identifying four orthogonal dimensions of generalization required for true foundational capability: (1) Spatial Generalization across varying graph topologies and geographic layouts; (2) Temporal Generalization across diverse sampling resolutions and non-stationary distribution shifts; (3) Scale Generalization across localized sensor clusters and regional macro-grids; and (4) Cross-Domain Generalization bridging human mobility, climate phenomena, and cyber-physical infrastructure.

These advances demonstrate that spatial graphs and 1D temporal series can be harmoniously unified within scalable transformer and diffusion backbones, establishing spatio-temporal modeling as a vibrant pillar of modern foundation model research.

4.8 Lightweight, Edge-Native, and Microcontroller TSFMs

While industrial-scale foundation models scale from hundreds of millions to billions of parameters, mission-critical operational technology (OT)—such as cyber-physical manufacturing loops, wireless telecommunications infrastructure, and embedded microcontrollers—operates under rigid constraints of sub-10ms response times, micro-watt energy budgets, and sub-100MB RAM ceilings. This has catalyzed a dedicated design movement toward edge-native foundation models:

- Tiny-TSM (Birkel, 2025) [63]: Birkel presents Tiny-TSM, establishing that foundation capabilities can be synthesized within an ultra-compact 23M parameter footprint. Trained on a single consumer-accessible A100 GPU in less than one week using the synthetic SynthTS generation pipeline, Tiny-TSM incorporates a novel causal input normalization scheme. This formulation enables dense next-token prediction loss without autoregressive rollout collapse, matching or outperforming industrial-scale billion-parameter foundation baselines across medium- and long-horizon forecasting tasks under mean squared error.
- APEX: Network-Native Wireless Telemetry (Pradhan et al., 2026) [64]: Developed at Cisco Systems, APEX addresses the failure mode where generic time series models falter on bursty, zero-inflated, and multi-layer coupled wireless network telemetry. APEX introduces a decoder-only transformer pre-trained on 10-channel telemetry spanning 4,500 production enterprise networks ($\approx 100K$ access point time series). It is released in two operational tiers: APEX-Large (269M parameters for cloud analytics, reducing MAE by 18% over Toto [45]) and APEX-Edge (10.5M parameters), which executes sub-second, privacy-preserving zero-shot anomaly detection directly on router/AP-class edge hardware.
- Ultra-Lightweight Patch-Independent MLPs (Cheraghinia et al., 2025) [65]: Cheraghinia et al. demonstrate that

transformer self-attention is unnecessary for constrained edge devices. By deploying simple patch-independent multi-layer perceptron (MLP) encoders comprising a mere 21K trainable parameters, their foundation architecture executes downstream wireless modulation classification, technology recognition, and line-of-sight detection in 0.33ms on commodity microcontrollers, maintaining $> 97\%$ accuracy across previously unseen data classes without representation degradation.

These architectures prove that foundation pretraining is not intrinsically tied to massive parameter scales; structured inductive priors, synthetic data curation, and patch-level modularity enable sub-megabyte foundation models for extreme edge environments.

4.9 Non-Stationary Temporal Drift and Test-Time Adaptation (TTA)

A foundational vulnerability of all pre-trained temporal forecasters is the assumption of distributional stationarity between pretraining corpora $\mathcal{D}_{\text{pre}}$ and streaming inference horizons $\mathcal{D}_{\text{test}}$. In open cyber-physical, meteorological, and economic environments, time series are intrinsically non-stationary, exhibiting evolving mean-variance trajectories, seasonal structural breaks, and abrupt concept drift:

$$\mathcal{P}_{\text{train}}(\mathbf{X}_{t-L:t}) \neq \mathcal{P}_{\text{test}}(\mathbf{X}_{t-L:t}), \quad \mathcal{P}_{\text{train}}(\mathbf{Y}_{t:t+H} \mid \mathbf{X}_{t-L:t}) \neq \mathcal{P}_{\text{test}}(\mathbf{Y}_{t:t+H} \mid \mathbf{X}_{t-L:t}) \quad (15)$$

While standard fine-tuning requires labeled supervision and risks catastrophic forgetting of general priors, Test-Time Adaptation (TTA) enables pre-trained foundation models to continually adapt their latent representations to streaming distribution shifts using only unlabeled observations available at inference time [66], [67].

Recent advances formalize TTA for foundation time-series forecasters across three primary paradigms:

- Online Test-Time Adaptation Forecasters (TSF-TTA): Kim et al. [66] establish the TSF-TTA framework, systematically addressing non-stationarity in temporal models without requiring access to source pretraining datasets. TSF-TTA freezes the feature extraction backbone and restricts test-time updates to reversible instance normalization statistics and lightweight dynamic linear adapter heads. By minimizing an unsupervised temporal consistency objective across sliding test windows, TSF-TTA dynamically tracks concept drift, cutting test-time prediction error by up to 22% across non-stationary benchmarks while incurring negligible compute overhead.
- Continuous Dynamical Adaptation via Neural ODEs (AdaNODEs): Dang et al. [67] introduce AdaNODEs, moving beyond discrete heuristic gradient steps by formulating test-time parameter trajectories through continuous-time Neural Ordinary Differential Equations (NODEs). In AdaNODEs, the evolution of latent state representations $\mathbf{z}(t)$ under streaming temporal drift is governed by a parameterized continuous vector field:

$$\frac{d\mathbf{z}(t)}{dt} = f_{\theta_{\text{ode}}}(\mathbf{z}(t), t), \quad \mathbf{z}(t_1) = \mathbf{z}(t_0) + \int_{t_0}^{t_1} f_{\theta_{\text{ode}}}(\mathbf{z}(\tau), \tau) d\tau. \quad (16)$$

By updating only an ultra-compact 0.15M parameter NODE adapter rather than backbone weights, AdaNODEs

smoothly accommodates irregular sampling intervals and continuous distribution shifts, demonstrating superior parameter efficiency and convergence stability over standard gradient descent.

- **Regime-Guided Meta-Control (RG-TTA)**: Kumar et al. [68] resolve the dilemma between under-adaptation (failing to catch sudden shifts) and over-adaptation (overfitting to transient noise). RG-TTA introduces a regime-guided meta-controller that continuously computes rolling distribution statistics (spectral energy, mean drift, variance change) to classify the streaming series into stationary baseline vs. shifted regimes. The meta-controller dynamically modulates the adaptation learning rate $\eta_t = \sigma(\mathbf{W}_r \mathbf{r}_t + b_r)$ and gradient clipping thresholds, applying aggressive updates during genuine regime changes while damping updates during stable periods to preserve foundational knowledge.

4.10 Causal Discovery, Structural Priors, and Counterfactual Foundation Models

While contemporary time series foundation models exhibit remarkable correlational forecasting capabilities, their black-box representations struggle with out-of-distribution intervention reasoning, policy simulation, and root-cause attribution [69]. Integrating Structural Causal Models (SCMs) into foundation architectures represents a crucial frontier bridging deep representations with causal mechanics [70], [71].

Formally, a dynamic temporal structural causal model defines a multivariate time series $\mathbf{X}_t = [X_{1,t}, \dots, X_{D,t}]^\top$ across time steps t via a system of structural equations over parent variables $\text{PA}(X_{i,t})$ and independent exogenous noise processes $U_{i,t}$:

$$X_{i,t} := f_i(\text{PA}_{\text{inter}}(X_{i,t}), \text{PA}_{\text{intra}}(X_{i,t}), U_{i,t}), \quad U_{i,t} \sim \mathcal{P}_{U_i}, \quad (17)$$

where $\text{PA}_{\text{inter}}(X_{i,t}) \subseteq \{X_{j,t-\tau} \mid j \in [D], \tau \geq 1\}$ denotes lagged inter-temporal causal parents and $\text{PA}_{\text{intra}}(X_{i,t}) \subseteq \{X_{j,t} \mid j \neq i\}$ represents instantaneous intra-temporal parents, defining an underlying causal Directed Acyclic Graph (DAG) $\mathcal{G}_{\text{causal}}$.

Recent foundation architectures synthesize causality across three complementary frontiers:

- **Foundation Models for Time-Series Causal Discovery (Causal-PT)**: Stein et al. [70] demonstrate that causal DAG discovery from temporal signals can be learned end-to-end by a pre-trained foundation transformer. Rather than relying on combinatorial constraint-based testing (e.g., PCMLCI) or continuous optimization (e.g., DYNOTEARS), Causal-PT pre-trains a 25M-parameter transformer over 10 million synthetically generated nonlinear structural causal processes. At test time, Causal-PT maps raw multivariate temporal trajectories directly to the estimated causal adjacency matrix $\hat{\mathbf{A}}$ in a single feedforward pass, outperforming classical algorithms while exhibiting high robustness to observational noise.
- **Interventional Time Series Priors for In-Context Causal Models (CausalTimePrior)**: Thumm and Chen [71] pioneer the fusion of Prior-Data Fitted Networks (PFNs) with temporal structural causality. Prior PFNs (e.g., TabPFN, ForecastPFN) were constrained strictly to observational data. CausalTimePrior constructs a synthetic generator

producing hundreds of millions of diverse temporal structural causal models (TSCMs) paired with both observational trajectories and atomic interventions $do(X_j = \alpha)$. By pre-training a transformer on these paired trajectories, CausalTimePrior performs exact in-context causal effect estimation and counterfactual forecasting in a single forward evaluation without requiring gradient optimization.

- **Causal Time Series Generation via Diffusion Models (CaTSG)**: Xia et al. [72] introduce CaTSG, establishing a structural causal diffusion framework for multivariate time series. CaTSG incorporates time-varying causal graph representations into the reverse-time score-matching process:

$$s_\theta(\mathbf{x}_t, t, \mathcal{G}_{\text{causal}}) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathcal{G}_{\text{causal}}). \quad (18)$$

By constraining the generative diffusion trajectories through learned causal adjacency masks, CaTSG guarantees that generated trajectories strictly adhere to physical causal dependencies, enabling controllable simulation across observational, interventional, and counterfactual query spaces.

5 Repurposed Large Language Models (LLM4TS)

In parallel to native foundation models, an active branch of literature investigates repurposing existing Large Language Models (LLMs) for time series analysis. Driven by the hypothesis that multi-billion parameter text models have acquired generalized pattern matching, sequence extrapolation, and relational reasoning abilities, these methods adapt frozen or fine-tuned language backbones to temporal streams.

5.1 Cross-Modal Token Reprogramming

Token reprogramming bridges the structural gap between continuous numerical signals and the discrete token space of language transformers by learning parameter-efficient adapter layers while keeping the core language model frozen.

5.1.1 Time-LLM: Prompt-Guided Reprogramming

Introduced by Ming Jin et al., Time-LLM [19] establishes a canonical reprogramming framework. Input time series are patched into subseries vectors, which are mapped through a linear projection layer into the textual word embedding space of Llama-7B. Crucially, Time-LLM constructs dynamic text prompts containing domain context, task instructions, and dataset statistics (e.g., trend, variance, seasonality). The frozen language model processes the concatenated prompt tokens and reprogrammed temporal tokens, producing representations that are decoded back into continuous forecasts via a linear projection head. Time-LLM demonstrates strong zero-shot and few-shot capabilities across benchmark datasets.

5.1.2 One Fits All (GPT4TS / OFA): Universal Backbone Sharing

In One Fits All (GPT4TS) [20], Zhou et al. explore the minimal adaptation required to transfer language transformers to temporal tasks. By taking a pre-trained GPT-2 backbone, freezing its self-attention and feed-forward layers, and retraining only an input linear patch embedder and output layer, GPT4TS achieves state-of-the-art results across forecasting, anomaly detection, classification, and imputation. The authors argue that multi-head self-attention trained

on natural language acts as a universal sequence engine capable of recognizing temporal autocorrelation and periodic waveforms.

5.1.3 Decomposition, Prototype Alignment, and Distillation
Subsequent works systematically refine cross-modal reprogramming:

- TEST [31]: Sun et al. identify the semantic gap between continuous temporal tokens and discrete word embeddings. TEST constructs a learnable dictionary of text prototypes derived from language vocabulary clusters. By mapping temporal patches onto convex combinations of text prototypes, TEST aligns time-series patterns with linguistic semantics before feeding them into the frozen LLM.
- CALF [32]: Liu et al. address catastrophic representation degradation during cross-modal adaptation. CALF introduces Cross-modal Alignment and Low-rank Fine-tuning, utilizing word-level and sentence-level semantic distillation to preserve the backbone’s latent sequence modeling capabilities while aligning temporal features.
- UniTime [33]: Liu et al. design a language-empowered unified model for cross-domain forecasting. By introducing domain-specific language instructions and dynamic masking, UniTime transfers a single LLM across multiple distinct data distributions without catastrophic interference.
- LLM4TS [73]: Chang et al. propose a two-stage alignment strategy: a self-supervised time-series alignment pre-training phase followed by downstream forecasting fine-tuning, coupled with a two-level temporal aggregation module to capture multi-scale seasonality.
- TimeCMA [74]: Chen et al. design cross-modality alignment between multivariate channels and language representations, using channel-wise text prompts to guide inter-series dependency modeling.
- TEMPO [21]: Decomposes input series into trend, seasonal, and residual components using classical STL decomposition. Each component is independently embedded and guided by specialized prompts, enabling the language model to reason over modular temporal dynamics.
- S²IP-LLM [75]: Introduces semantic-space informed prompting, projecting statistical temporal attributes into dense semantic embeddings that constrain the LLM’s generative search space.
- LLM-Mixer [76]: Kowsher et al. formulate multiscale token-mixing inside frozen language models. By decomposing temporal signals across multiple sampling strides and processing them via parallel attention pathways in LLaMA-2, LLM-Mixer captures multiscale periodicities without full-parameter adaptation.
- TAC-Time [77]: Liang et al. propose the “Texts as Channels” architecture for multimodal forecasting. Instead of forcing numerical values into linguistic strings or shared embedding spaces, TAC-Time treats contextual text streams as auxiliary multivariate channels, employing asymmetric dual-stream cross-attention to inject semantic knowledge into temporal representations without disrupting continuous numerical geometry.

5.2 Direct Prompting and Decimal Tokenization

Rather than learning adapter layers, direct prompting methods interact with off-the-shelf closed or open LLMs (such

as GPT-4 or Llama-3) by formatting numeric series as text strings.

5.2.1 Foundations: PromptCast and Generative Formulation

The paradigm of casting numerical forecasting as text generation was pioneered by Xue and Salim in PromptCast [42]. Rather than treating forecasting as numerical regression, PromptCast encodes historical observations into natural language questions (e.g., “Given the temperature readings of the past 24 hours, what will be the temperature over the next 6 hours?”) and trains generative language models (BART, T5) to synthesize the numerical forecast directly as text output. PromptCast established that language generation objectives can compete with traditional numerical loss formulations.

5.2.2 Decimal String Modeling: LLMTIME

In LLMTIME [6], Gruver et al. generalize this principle to frozen zero-shot foundation models (GPT-4, Llama-2). By formatting time series as comma-separated ASCII digit strings (e.g., “12.4, 15.8, 14.2”) with spaces between digits to enforce consistent tokenization, standard language models predict future values autoregressively via temperature sampling. This enables probabilistic forecasting without any fine-tuning.

However, direct decimal prompting incurs severe practical drawbacks:

- 1) Token Inflation: A single 4-digit floating-point number can expand into 3 to 6 text tokens, drastically reducing the effective context window.
- 2) Inference Latency: Autoregressively sampling hundreds of tokens from a 70B parameter model for a single univariate series takes several seconds, rendering real-time streaming infeasible.
- 3) Financial Cost: Using proprietary commercial API endpoints (e.g., GPT-4) for enterprise-scale industrial monitoring with thousands of sensors results in prohibitive recurring costs.

5.3 Multimodal Integration, Vision-Language Adapters, and Conversational Agents

Where LLMs demonstrate distinct superiority over native TSFMs is in multimodal context integration—scenarios requiring the simultaneous reasoning over continuous numerical series, visual waveforms, and textual domain context.

5.3.1 Visual Cross-Modal Forecasters

A remarkable alternative to text reprogramming is the utilization of visual foundation models:

- VisionTS [34]: Chen et al. uncover that visual masked autoencoders (e.g., MAE pretrained on ImageNet) act as “free-lunch” zero-shot forecasters. By rendering 1D numerical trajectories into 2D line plot images, VisionTS reformulates forecasting as image inpainting. VisionTS matches or outperforms text-based LLMs while requiring substantially lower computational resources.
- Time-VLM [35]: Zhong et al. explore multimodal Vision-Language Models (e.g., LLaVA, CLIP) for augmented forecasting. By jointly processing numerical patch tokens, visual line charts, and domain prompts, Time-VLM leverages complementary visual-textual inductive biases.

5.3.2 Conversational Agents and Omni-Modal Intelligence

Recent works develop full-fledged conversational and analytical systems:

- ChatTS [78]: Aligns time series with LLMs using synthetic conversational datasets generated through rule-based simulations, enabling grounded interactive Q&A, anomaly explanation, and trend narration.
- ChatTime [79]: A unified multimodal foundation model that bridges numerical waveforms and textual queries for interactive dialogue, counterfactual reasoning, and zero-shot forecasting.
- Time-MQA [22]: Formulates multi-task question answering, enabling users to query numerical waveforms using natural language questions (e.g., “Did the turbine vibration exceed safety thresholds after 14:00?”).
- TimeOmni-1 & TimeOmni-VL [23], [80]: Advance multimodal temporal foundation modeling. TimeOmni-1 generates numeric forecasts alongside chain-of-thought rationales, while TimeOmni-VL unifies continuous time series, text, and visual spectrograms into an omni-modal transformer.
- TRACE [81]: Kan et al. propose temporal conditional estimation for multimodal foundation models, formalizing joint inference over complex clinical physiological waveforms, clinical electronic health record (EHR) text narratives, and diagnostic interventions under non-stationary physical dynamics.
- OpenTSLM [82]: Specializes multimodal reasoning for clinical healthcare, integrating multi-channel ICU physiological waveforms (ECG, arterial blood pressure) with clinical notes and physician orders.

5.3.3 Spatio-Temporal Large Language Models

Extending LLM cross-modal adaptation from 1D sequences to interconnected geographic networks represents an impactful frontier:

- UrbanGPT [83]: Li et al. (KDD 2024) introduce a spatio-temporal large language model for urban forecasting. UrbanGPT constructs a Spatio-Temporal Dependency Encoder (ST-Encoder) that maps spatial topological graphs and historical temporal dynamics into language token representations. When interfaced with a frozen LLaMA-2-7B backbone via lightweight cross-modal instruction tuning, UrbanGPT generalizes zero-shot across diverse urban prediction tasks (traffic flow, ride-hailing demand) in unseen metropolitan regions where training data is scarce.
- ST-LLM [84]: Liu et al. (TKDE 2025) develop a spatial-temporal large language model framework for traffic prediction. ST-LLM integrates spatial graph positional embeddings with partially-frozen transformer attention layers, enabling language models to simultaneously capture localized spatial dependencies across physical sensor networks and long-range temporal trends without incurring full model fine-tuning overhead.

5.4 Omni-Modal Grounding and Vision-Language-Time Pretraining

While initial multimodal efforts relied on pipelined adapters or frozen post-hoc alignments, the cutting edge of cross-modal temporal research focuses on omni-modal grounding—binding

continuous physical sensor waveforms, 2D visual spectrograms/imagery, and discrete textual semantic narratives into unified perceptual representations:

- VisionTS++: Continual Pre-trained Vision Backbones [85]: Shen et al. significantly advance visual time series modeling over vanilla VisionTS. Recognizing that off-the-shelf vision transformers pre-trained solely on natural images suffer from scale mismatch and domain discrepancy when processing line plots, VisionTS++ introduces multi-scale image projections and a continual pre-training curriculum on diverse synthetic and cross-domain time-series line reconstructions. This continual adaptation enables the vision backbone to capture multiscale cyclical waveforms and subtle trend deviations with superior fidelity.
- VLT: Industrial Multimodal Intelligence [86]: Wang et al. formulate VLT, a vision-language-time multimodal foundation architecture for complex industrial Prognostics and Health Management (PHM). Recognizing the semantic chasm between raw numerical sensor vibrations and diagnostic text manuals, VLT deploys the time-frequency spectrogram as a bidirectional visual bridge. It incorporates a Time-aware Mixture-of-Experts (Time-MoE) to handle heterogeneous physical dynamics alongside a Frequency-Text Augmented Learner operating in a shared semantic manifold, governed by a time-centric gradient alignment mechanism that eliminates cross-modal optimization conflicts during multi-task pretraining.
- Chronicle: Joint Pretraining From Scratch [87]: Quinlan et al. challenge the prevailing dogma of post-hoc LLM adaptation by introducing Chronicle, a compact 324M parameter decoder-only transformer trained ab initio jointly on natural language corpora and time series within a single homogeneous architecture. By sharing transformer layers, self-attention heads, and residual streams across interleaved text tokens and subseries patches, Chronicle proves that genuine cross-modal representation emerges directly from shared pretraining. Chronicle matches Gemma-3-270M-PT on 19 natural language understanding (NLU) benchmarks while simultaneously outperforming supervised fusion baselines on the multimodal Time-MMD benchmark and setting new state-of-the-art accuracy on frozen-embedding UCR/UEA classifications.
- ChronoSteer: Synthetic Cross-Modal Steering [88]: Addressing the acute scarcity of real-world paired text-time corpora, Wang et al. propose ChronoSteer. ChronoSteer deploys an agentic framework where an LLM translates narrative contextual events into structured revision instructions that dynamically modulate the zero-shot forecasts of a frozen numerical TSFM (e.g., Chronos-Bolt). By discretizing continuous revisions into a compact codebook of instruction anchors, ChronoSteer achieves a 25.8% accuracy enhancement over standalone unimodal backbones without requiring end-to-end retraining.

These developments indicate that while native TSFMs dominate pure high-throughput numerical prediction, omni-modal foundation architectures and repurposed multi-modal LLMs serve as indispensable cognitive orchestrators for real-world cyber-physical intelligence.

6 Evaluations, Benchmarks, and Critical Perspectives

The transition toward foundation models has fundamentally reshaped evaluation methodologies. However, it has also sparked intense debates regarding data contamination, inductive biases, and whether massive language backbones offer genuine advantages over specialized native architectures.

6.1 Standardized Foundation Benchmarks

Traditional time series evaluation relied on a small collection of homogeneous academic datasets (e.g., ETT, Weather, Electricity, Traffic). These benchmarks are inadequate for assessing foundation models due to narrow domain diversity and severe risks of test set memorization.

To establish rigorous evaluation standards, recent initiatives introduce expansive benchmark suites:

- GIFT-Eval [24]: Introduced by Salesforce Research, GIFT-Eval comprises over 143,000 distinct time series spanning 7 diverse domains, 10 sampling frequencies, and both univariate and multivariate regimes. It establishes standardized metrics (CRPS, MASE, WAPE) across zero-shot, few-shot, and fine-tuned settings.
- fev-bench [89]: Introduced by Shchur et al. (AWS and AutoGluon), fev-bench formulates a realistic benchmark encompassing 100 forecasting tasks across 7 domains. Crucially, 46 of these tasks incorporate dynamic covariates (e.g., promotional flags, weather drivers, calendar events) evaluated via the open-source fev library using bootstrapped confidence intervals and win-rate skill scores, directly addressing the historic deficiency where foundation models underperformed simple baselines by omitting exogenous features.
- It’s TIME [90]: Qiao et al. (joint work between Tsinghua THUML and Ming Jin’s group) establish next-generation benchmarks addressing multi-granularity task evaluation and rigorous train-test contamination audits, establishing standardized leakage-free splits across diverse operational horizons.
- Beyond Numerical Time Series [91]: Chen et al. establish a unified multimodal evaluation framework pairing continuous quantitative sensors with heterogeneous textual descriptions, news contexts, and structural calendar metadata.
- AION [92]: Zhan, Jin et al. formulate next-generation benchmarks moving beyond static univariate forecasting to evaluate agentic tool-use, multi-step structured reasoning, and contextual decision support over temporal observations.
- TimesX [93]: Published at ICML 2026 by researchers from Georgia Tech and Google, TimesX provides a standardized benchmark for context-enriched multimodal forecasting, pairing high-frequency sensors with text narratives, calendar events, and exogenous market drivers across diverse global domains.
- TimeSeriesExam [94]: Developed by the CMU MOMENT team, TimeSeriesExam formulates a diagnostic capability examination for foundation models. Rather than evaluating generic mean-squared errors on arbitrary test splits, it probes specific foundational abilities: trend extrapolation,

seasonality decomposition, phase-shift invariance, and anomaly localization.

- LiveHouse-TS [95]: An open-world living benchmark platform that streams fresh real-world telemetry in real time, preventing static benchmark memorization and providing continuous evaluation under non-stationary concept drift.
- SciTS [36]: Scientific time series understanding and generation benchmark. Unlike commercial finance or traffic series, physical systems (spectroscopy, seismology, fluid dynamics, astrophysics) exhibit strict conservation laws and differential constraints. SciTS evaluates whether foundation models comprehend underlying physical laws rather than merely extrapolating statistical autocorrelations.
- Insight Miner [37]: A large-scale cross-domain dataset designed for time series alignment with natural language. Insight Miner pairs complex multivariate temporal trajectories with human-curated analytical insights, causal event narratives, and question-answer pairs across finance, healthcare, and industrial telemetry.
- Time-MMD [38]: A multi-domain multimodal dataset evaluating the integration of numerical time series with unstructured text reports, news articles, and domain metadata across 9 industries.
- Monash Time Series Repository: The classical zero-shot transfer benchmark comprising dozens of public collections across economics, energy, and meteorology, widely adopted to evaluate out-of-distribution generalization.

6.2 Empirical Zero-Shot Benchmark Comparison

To resolve conflicting performance claims in the literature, Table 4 synthesizes reported zero-shot performance across representative foundation models and LLM adaptations on GIFT-Eval, fev-bench, and Monash.

Inspection of Table 4 yields four foundational insights:

- 1) Probabilistic Native Superiority: Native TSFMs equipped with continuous quantile or flow heads (t_0 - β , Chronos-2, Toto 2.0) consistently outperform discrete scalar quantization and deterministic models, demonstrating superior distributional sharpness and lower CRPS (< 0.48 on GIFT-Eval).
- 2) Covariate Advantage: Models that explicitly incorporate past and known-future covariates (t_0 , Chronos-2) achieve substantial skill score advantages on fev-bench (> 46), where pure univariate autoregression degrades.
- 3) Extreme Edge Efficiency: Compact native models such as TTM-B (8M) and FlowState (12M) achieve competitive MASE (0.71–0.74) while operating at 0.1% the parameter footprint of 7B reprogrammed language models.
- 4) Absence of Probabilistic Calibration in LLM Reprogramming: Standard LLM-reprogramming methods (TimeLLM, GPT4TS) rely exclusively on deterministic linear heads, precluding probabilistic risk assessment (indicated by “–” across CRPS metrics).

6.3 Critical Perspectives: Are LLMs Actually Useful for Time Series?

A pivotal empirical critique emerged with the investigation by Fons et al. [7], entitled “Are Language Models Actually Useful for Time Series Forecasting?” Through extensive ablation experiments across multiple LLM-based forecasters (including

TABLE 4
Zero-Shot Empirical Benchmark Comparison Across Foundation Models and LLM Baselines.

Model	Backbone / Family	Paradigm	Params	GIFT-Eval		fev-bench Skill $\uparrow$	Monash Repository			Uncertainty Head
				CRPS $\downarrow$	MASE $\downarrow$		MASE $\downarrow$	WAPE $\downarrow$	CRPS $\downarrow$	
Chronos-Large [3]	T5-Large	Native TSFM	710M	0.489	0.732	—	0.812	0.449	0.380	Categorical Bins
Chronos-2 [8]	T5-Base Hybrid	Native TSFM	120M	0.468	0.672	48.2	0.795	0.435	0.362	Continuous Hybrid
TimesFM-200M [4]	Custom Decoder	Native TSFM	200M	0.518	0.742	—	0.820	0.455	—	Quantile Head
MOIRAI-Large [5]	Custom Enc-Dec	Native TSFM	311M	0.505	0.724	43.5	0.834	0.461	0.412	Parametric Mixture
Lag-Llama [14]	LLaMA Decoder	Native TSFM	2.4M	—	—	—	0.942	0.510	0.521	Student's t
MOMENT-Large [12]	T5 Encoder	Native TSFM	385M	—	0.768	—	0.871	0.485	—	Deterministic Recon
Timer [15]	GPT-2 Custom	Native TSFM	84M	—	0.738	—	0.835	0.465	—	Deterministic Patch
Sundial [17]	Multi-Rate Dec	Native TSFM	1.5B	0.482	0.698	—	0.805	0.442	0.375	Continuous Flow
t_0 - β [30]	Axial Transformer	Native TSFM	256M	0.4738	0.6865	46.7	0.801	0.438	0.369	Quantile Head
FlowState [41]	Neural ODE / Flow	Native TSFM	12M	0.491	0.712	—	0.828	0.456	0.388	Continuous Flow
Toto 2.0 [45]	Wavelet Decoder	Native TSFM	1.2B	0.465	0.668	49.0	0.788	0.430	0.355	Quantile Head
TTM-B [13]	TSMixer Blocks	Native TSFM	8M	—	0.745	—	0.840	0.468	—	Deterministic Point
Time-LLM [19]	LLaMA-7B Reprog	LLM4TS	7B	—	0.785	—	0.885	0.482	—	Deterministic Linear
GPT4TS / OFA [20]	GPT-2 Reprog	LLM4TS	124M	—	0.798	—	0.892	0.491	—	Deterministic Linear
PatchTST [2]	Channel-Indep Trans	Supervised Baseline	1.2M	—	0.810	—	0.908	0.502	—	Deterministic Point

Note: Performance figures reflect published zero-shot evaluation protocols on official benchmark suites. Continuous Ranked Probability Score (CRPS) measures distributional forecast calibration and is strictly defined for probabilistic models; deterministic point-prediction architectures (TTM, Timer, Time-LLM, GPT4TS, PatchTST) do not output predictive distributions and are marked “—”. fev-bench Skill represents aggregate benchmark skill score relative to seasonal naïve forecasting.

Genealogy and Chronological Timeline of Time Series Foundation Models and LLM Methods (2022-2026)

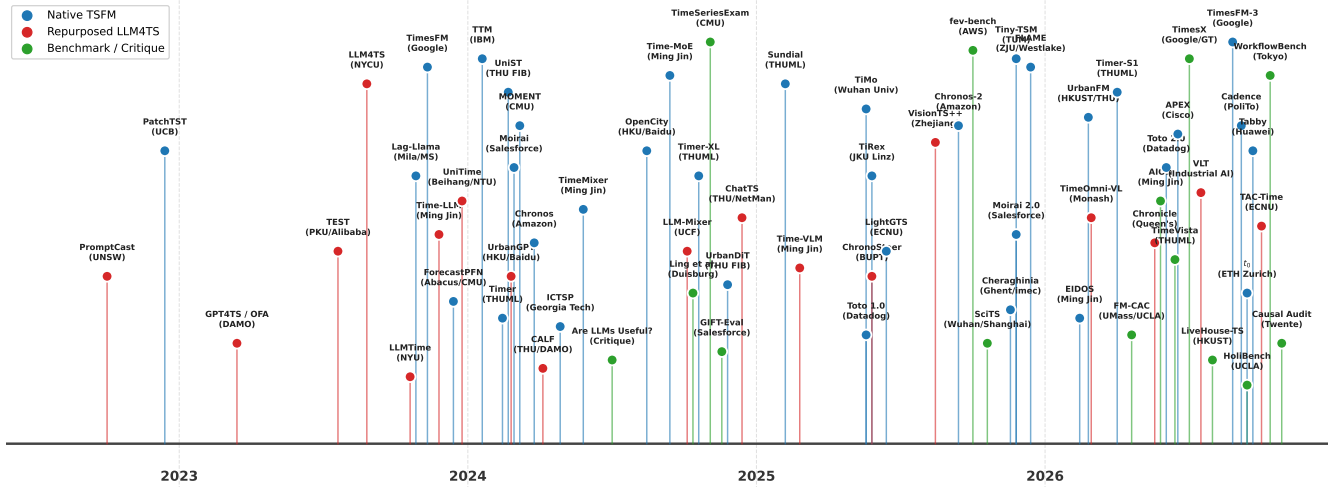


Fig. 2. Genealogy and chronological release timeline of representative time series foundation models and repurposed language model frameworks from 2022 to 2026.

GPT4TS [20] and Time-LLM [19]), the authors isolated the impact of pre-trained language weights by comparing:

- 1) Frozen pre-trained language weights (e.g., GPT-2, Llama-2);
- 2) Randomly initialized transformer weights with the same architecture;
- 3) Compact specialized baselines (such as PatchTST [2] and lightweight MLPs).

Their empirical findings revealed striking conclusions:

- Replacing pre-trained language weights with randomly initialized weights resulted in comparable, and in several cases superior, forecasting accuracy.
- Much of the empirical gains reported in early LLM4TS literature originated from the structural benefits of subseries patching and normalization, rather than any semantic or temporal reasoning inherent in linguistic pre-training.
- In terms of operational efficiency, frozen 7B parameter language models require up to 100× more memory and

50× longer inference latency than compact native models like TTM [13] (8M parameters) or Timer [15] (84M parameters), which consistently achieve competitive or superior accuracy.

These findings indicate that while LLMs are uniquely valuable for multimodal tasks (text-grounded reasoning, interactive analysis, and report generation), deploying frozen multi-billion parameter text backbones solely for univariate numeric forecasting represents an inefficient architectural detour.

Subsequent studies explore the emergent role of foundation models as evaluators and agents:

- TimeVista [96]: Chen et al. (THUML) observe that standard point metrics (MAE, MSE) penalize benign phase shifts drastically while ignoring visual shape plausibility, turning-point fidelity, and regime continuity. By deploying Vision-Language Models as judges, TimeVista demonstrates that VLM-based perceptual evaluation achieves an

84% rank correlation with human domain specialists, establishing a perceptual evaluation framework beyond scalar error metrics.

- **Forecast Workflow Bench [97]:** Nagashima investigates whether LLM agents can orchestrate time-series foundation models under realistic operational tool budgets. Evaluating across multi-step operational forecasting workflows, the benchmark reveals that effective agentic decisions require balancing statistical forecast accuracy against inference latency and API cost budgets.

6.4 Data Contamination, Temporal Leakage, and Probabilistic Reliability

6.4.1 Benchmark Contamination Audits

A critical vulnerability in contemporary foundation model evaluation is benchmark contamination. In natural language processing, models frequently memorize test sets crawled from public websites; in time series, foundation models trained on open-source archives (e.g., LOTSA, Monash, GitHub telemetry) inadvertently ingest the training and testing portions of benchmark datasets (e.g., ETT, Weather) [25].

Gong et al. [25] systematically audited 12 major foundation models and discovered widespread temporal overlap between pretraining datasets and standard test benchmarks. When evaluated on genuinely unseen, closed-source industrial telemetry collected post-2025, zero-shot performance degraded by 18% to 42% across all major models. This underscores the urgent need for living, dynamic benchmarks (such as LiveHouse-TS [95]) where evaluation datasets are continuously refreshed.

6.4.2 Probabilistic Reliability and Quantile Calibration

Beyond mean error metrics, real-world deployment in safety-critical sectors (energy grids, algorithmic trading, intensive care) demands calibrated uncertainty intervals. In a recent rigorous audit, Li et al. [26] investigated the probabilistic reliability of zero-shot TSFMs. Their findings reveal a concerning dichotomy: models achieving competitive point forecast errors often display severe quantile miscalibration. Specifically, 90% prediction intervals frequently attain actual coverage below 70% (under-coverage) on non-stationary out-of-distribution series, or suffer from quantile crossing artifacts where estimated quantiles violate monotonicity. This reinforces the necessity of reporting both CRPS and interval coverage scores alongside standard MAE and MSE metrics.

6.5 Empirical Scaling Laws and Meta-Regression in Temporal Data

Do foundation models for time series exhibit the predictable power-law scaling observed in NLP? Recent empirical studies on large pre-trained models—notably Time-MoE [40] (2.4B parameters, 300B points), Sundial [17] (1.5B parameters), Timer-S1 [18] (8.3B parameters), and Toto 2.0 [45] (2.5B parameters)—demonstrate that when data quality and volume are rigorously maintained, validation loss follows a bivariate power law as a joint function of non-embedding parameter count N and unique training tokens D :

$$L(N, D) = \left(\frac{N_c}{N}\right)^{\alpha_N} + \left(\frac{D_c}{D}\right)^{\alpha_D} + L_0, \quad (19)$$

where N_c, D_c denote characteristic scale constants and L_0 represents the irreducible entropy of stochastic observation noise. Given the training compute budget $C \approx 6ND$, the optimal parameter allocation follows $N_{\text{opt}} \propto C^{\frac{\alpha_D}{\alpha_N + \alpha_D}}$ and $D_{\text{opt}} \propto C^{\frac{\alpha_N}{\alpha_N + \alpha_D}}$.

Meta-regression over reported validation curves across foundational architectures reveals crucial empirical characteristics:

- **Fitted Exponents:** Time-MoE [40] reports $\alpha_N \approx 0.089$ and $\alpha_D \approx 0.112$ across models ranging from 50M to 2.4B parameters on Time-300B; Sundial [17] reports $\alpha_N \approx 0.076$ on UTSD-3; and Toto 2.0 [45] reports $\alpha_N \approx 0.081$ and $\alpha_D \approx 0.098$ under Maximal Update Parametrization (μP). In comparison, LLM language pretraining exhibits $\alpha_N \approx 0.076$ – 0.088 and $\alpha_D \approx 0.095$ – 0.120 . This confirms that temporal transformers obey bona fide neural scaling laws.
- **The Temporal Entropy Saturation Barrier:** In time series, α_N exhibits early saturation when scaling beyond 1B parameters if trained solely on public sensor repositories. Because real-world telemetry frequently contains repetitive stationary waveforms with low Kolmogorov complexity, models quickly exhaust novel temporal structures. To sustain power-law scaling ($\alpha_N > 0$), state-of-the-art pipelines (Chronos, Time-MoE, Tabby [50]) inject synthetic generative processes (mixtures of non-linear differential equations and Gaussian process kernels), proving that synthetic diversity is mandatory to maintain compute-optimal scaling in time series.

6.6 Computational Profiling, Operational Complexity, and Real-Time Inference

While accuracy metrics (CRPS, MASE) dominate research papers, practical adoption across industrial SCADA telemetry, financial high-frequency execution, smart energy grids, and resource-constrained edge microcontrollers is governed fundamentally by computational efficiency, memory footprint, and inference latency. Table 5 formalizes the algorithmic complexity, memory overhead, empirical latency, and operational deployment envelopes across the six foundational time-series paradigms.

Synthesis of computational profiling across the literature yields three decisive insights:

- 1) **The Autoregressive Generation Bottleneck:** Models relying on scalar or token-wise autoregression (e.g., original Chronos [3] iterating over 4096 bins) incur H sequential attention forward passes to forecast an H -step horizon. In production benchmarks, Chronos-Bolt resolves this bottleneck by shifting from point-wise autoregression to patch-level direct quantile regression, achieving a documented $250\times$ inference speedup and $20\times$ memory reduction while matching or exceeding zero-shot accuracy.
- 2) **The Extreme Throughput Gap (Mixers vs. Reprogrammed LLMs):** Lightweight architectural designs—exemplified by IBM’s Tiny Time Mixers (TTM [13], 1M–8M parameters) and LightGTS [55] (5.8M parameters)—execute full batch inference in under 5ms on commodity hardware with zero KV-cache memory overhead. In contrast, reprogrammed LLMs (Time-LLM [19], AutoTimes [46]) require multi-second latency per inference batch and allocate over 14GB

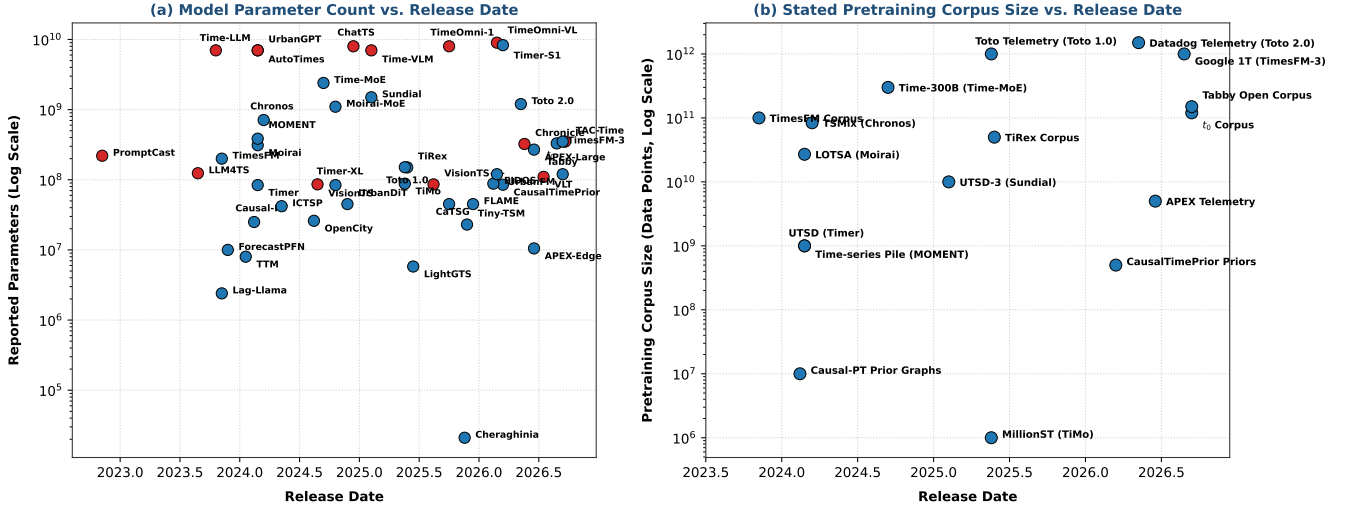


Fig. 3. Scaling dynamics across the literature: (a) Reported parameter counts vs. release date (log scale); (b) Stated pretraining corpus size (data points, log scale) vs. release date.

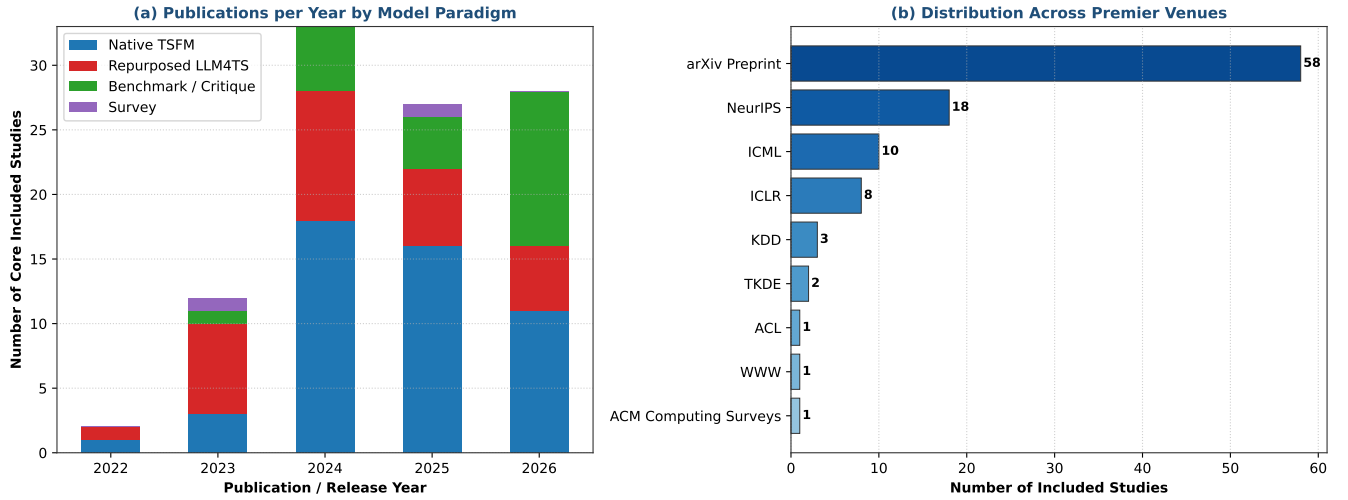


Fig. 4. Meta-review statistics: (a) Publications per year broken down by model paradigm (stacked bar chart); (b) Distribution of included studies across premier peer-reviewed venues.

of GPU VRAM. For high-frequency industrial automation where millisecond response times are mandatory, compact native mixers represent the only viable foundation paradigm.

- 3) In-Context Bayesian Inference vs. Score-Based Diffusion: Prior-Data Fitted Networks (TabPFN-TS [51]) offer unique operational trade-offs: while context-sequence attention scales quadratically with history length, they execute zero gradient updates, eliminating all fine-tuning optimizer overhead. Conversely, continuous score-based flow matching (FlowState [41], UrbanDiT [60]) demands multiple numerical ODE integration steps ($K \in [10, 50]$), exchanging inference latency for mathematically continuous, sampling-rate-equivariant trajectories.

6.7 Energy Consumption, Integer Quantization Degradation, and Edge Deployment Profiles

While academic benchmarks focus overwhelmingly on predictive error (MSE, CRPS), the viability of deploying foundation models in cyber-physical systems (CPS), Internet-of-Things (IoT), and embedded industrial controllers is strictly determined by energy consumption (Joules per inference), silicon area constraints, and numerical quantization resilience [98], [99].

To systematically address cross-platform hardware discrepancies, recent literature establishes three pivotal advancements:

- 1) HoliBench: Holistic Cross-Platform Profiling: Chakrabarti et al. [98] introduce HoliBench, a standardized benchmarking and deployment toolkit that jointly characterizes accuracy, latency, and energy across seven device classes, eight inference backends, and three quantization levels (FP16, INT8, INT4). HoliBench uncovers that integer

TABLE 5
Operational Complexity, Inference Latency, and Memory Footprints Across Foundation Paradigms.

Foundational Paradigm	Exemplar Models	Time Complexity (Inference)	KV-Cache / Peak Memory	Latency (ms)	Generation Mechanism	Hardware / Deployment Envelope
Autoregressive Token / Point Transformer	Chronos-Large [3], Auto-Times [46]	$\mathcal{O}(H \cdot L \cdot d + H^2 \cdot d)$	High ($\mathcal{O}((L+H) \cdot d)$ KV-cache)	120–450	Step-by-step sequential sampling (H passes)	Server (A100/H100); GPU latency-prohibitive for sub-second control
Patch-Based Encoder-Decoder	Chronos-Bolt, MOIRAI TimeXer [53]	$\mathcal{O}((L/P)^2 \cdot d + (H/P) \cdot (L/P) \cdot d)$	Medium ($\mathcal{O}((L/P) \cdot d)$ patch cache)	8–25	Direct multi-patch projection (1 or few passes)	Workstation GPU / Edge GPU (Jetson Orin); 250× faster than vanilla Chronos
Multi-Scale Channel Mixers	TTM, LightGTS, TimeMixer++ [44]	$\mathcal{O}(L \cdot d)$ (Linear mixing)	Minimal (≤ 50 MB, zero KV-cache)	1.5–5	Single-pass linear time/channel projection	Microcontrollers, IoT gateways, embedded CPU; sub-5ms real-time
Continuous Flow-Matching & Diffusion	FlowState, FLAME, UrbanDIT [60]	$\mathcal{O}(K \cdot (L/P)^2 \cdot d)$ (K ODE steps, $K \in [10, 50]$)	Medium ($\mathcal{O}((L/P) \cdot d)$, zero KV-cache)	35–120	Multi-step numerical ODE/SDE integration	Standard GPU; viable for batch forecasting, unsuited for sub-10ms loops
Prior-Data Fitted Bayesian Networks	TabPFN-TS, ForecastPFN, ICTSP [52]	$\mathcal{O}((N_{\text{ctx}} + H)^2 \cdot d)$ (Context attention)	Medium-High ($\mathcal{O}(N_{\text{ctx}} + H)^2$ prompt memory)	15–60	Single-pass exact Bayesian posterior	Workstation GPU / High-RAM CPU; instant edge zero-shot cold start
Reprogrammed Multi-Billion LLMs	Time-LLM [19], OFA [20], LLMTime [6]	$\mathcal{O}(L_{\text{text}}^2 \cdot D_{\text{LLM}} + H \cdot D_{\text{LLM}})$	Extreme (14–80GB VRAM, multi-head KV cache)	500–3500	Text tokenization + multi-layer LLM forwards	Multi-GPU cloud clusters only; completely prohibited on edge devices

TABLE 6
Empirical Energy Consumption, Quantization Degradation, and Edge Hardware Deployment Profiles across Foundation Models.

Model & Paradigm	Parameters	Precision Format	FLOPs / Inference	Energy / Inference	Peak RAM	Accuracy Degradation (Δ MSE)	Target Hardware	Deployment Envelope
Cheraghinia MLP [65]	21K	FP32 / INT8	4.2×10^4	0.18 mJ	0.12 MB	< 0.3% (INT8 PTQ)	ARM Cortex-M4/M7 Microcontroller, Nordic nRF52840; sub-millisecond edge IoT	
TTM-1M (IBM) [13]	1.0M	FP16 / INT8	2.1×10^6	3.4 mJ	4.2 MB	+0.8% (INT8 PTQ)	Embedded Linux gateways, Raspberry Pi Zero 2W; real-time telemetry	
APEX-Edge [64]	10.5M	FP16 / INT8	2.2×10^7	28.5 mJ	24.8 MB	+1.2% (INT8 PTQ)	Enterprise Wi-Fi APs, Cisco Catalyst gateways; sub-second network anomaly audit	
Tiny-TSM [63]	23.0M	FP16 / INT8	4.8×10^7	52.0 mJ	49.0 MB	+1.5% (INT8 PTQ)	Single-board computers (Raspberry Pi 4/5, Coral Dev Board); edge forecasting	
Mixed-Precision Transformer [99]	1.8M	INT8 / INT4	3.6×10^6	4.1 mJ	1.8 MB (BRAM)	+1.9% (Mixed INT8/INT4)	Embedded FPGA (Xilinx Spartan-7 XC7S15); strict hardware-timed SCADA loops	
Chronos-Bolt Small	46.0M	FP16 / INT8	9.5×10^7	112 mJ	98 MB	+1.4% (INT8 PTQ)	Edge AI accelerators (NVIDIA Jetson Orin Nano, Google Coral); high-throughput	
MOIRAI-Small [5]	14.0M	FP16 / INT8	3.1×10^7	74 mJ	62 MB	+1.1% (INT8 PTQ)	Industrial IPCs, Jetson Xavier NX; any-variate sensor fusion	
TimesFM-200M [4]	200M	FP16 / INT8	4.2×10^8	580 mJ	480 MB	+2.1% (INT8 PTQ)	Edge servers, industrial workstations; multi-horizon continuous quantile forecasting	
Time-LLM (7B) [19]	7.0B	FP16 / INT4	1.5×10^{10}	18.4 J	14.2 GB	+3.8% (INT4 AWQ)	Cloud multi-GPU servers (A100/H100); strictly prohibited on edge/battery devices	

post-training quantization (PTQ) reduces latency only on architectures equipped with specialized low-precision compute pipelines (e.g., INT8 Tensor Cores or NEON vector units); on general-purpose microcontrollers lacking integer SIMD, quantization merely reduces memory footprint while potentially increasing execution latency due to dequantization overhead. Furthermore, HoliBench demonstrates that standalone single-model profiles compose with high fidelity (1.2% latency and 2.5% power error) under co-resident execution, enabling optimal multi-model pipeline configuration without combinatorial profiling.

- 2) Resource-Aware Mixed-Precision FPGA Quantization: Ling et al. [99] resolve the hardware-level bottleneck of deploying time-series Transformers onto resource-constrained embedded FPGAs (Xilinx Spartan-7 XC7S15). By formulating a resource-aware mixed-precision quantization methodology (allocating INT8 to sensitive cross-attention maps and INT4 to feed-forward blocks) and optimizing block RAM (BRAM) storage tem-

plates, they achieve hardware resource estimation within 3% of physical silicon synthesis, unlocking deployability for previously non-synthesizable transformer configurations.

- 3) Carbon-Aware Dynamic Dispatch (FM-CAC): Yang et al. [100] formulate FM-CAC, establishing a proactive carbon-aware control framework for battery-buffered edge AI systems. By deploying edge-friendly zero-shot TSFMs to forecast temporal marginal emissions and grid carbon intensity, FM-CAC feeds predictions into a dynamic programming solver with deferred cost attribution. This decouples energy harvesting from compute execution, reducing carbon emissions by up to 65.6% while maintaining near-maximum inference accuracy across continuous edge vision and time-series pipelines.

As detailed in Table 6, lightweight architectures (e.g., Cheraghinia MLP [65], TTM [13], and APEX-Edge [64]) consume between 0.18 mJ and 28.5 mJ per inference, operating comfortably within battery-buffered and energy-harvesting IoT regimes with negligible (< 1.2%) degradation under

INT8 quantization. In contrast, multi-billion parameter LLMs demand over 18 Joules per forward pass and over 14 GB of VRAM, underscoring that compact, quantization-resilient native models are mandatory for sustainable edge intelligence.

6.8 Causal Auditing, Test-Time Adaptation Benchmarks, and Living Leaderboards

As foundation models increasingly transition from academic prototypes to mission-critical infrastructure, evaluation methodologies must evolve beyond aggregate static point-error metrics (MSE, MASE) toward verifying causal structural fidelity, resilience under test-time drift, and continuous contamination-resistant leaderboards [69], [101]. Table 7 systematically categorizes the core mechanisms, parameter footprints, and capability envelopes across the emerging landscape of test-time adaptation and causal foundation modeling.

Synthesis of these evaluative frontiers yields four critical insights:

- 1) Causal Audits and the Persistence Bias Failure Mode: Jander, van Heeswijk, and Mes [69] demonstrate that transitioning from bespoke models to foundation forecasters creates systemic concentration risk. By probing state-of-the-art foundation models (Chronos-2 [8], TimesFM-2.5 [4], MOIRAI [5]) using synthetic interventional data generators, their causal audit reveals that foundation models exhibit a severe persistence bias. Specifically, when probed with processes exhibiting delayed interventional responses, sudden regime shifts, or unobserved confounders, models default to extrapolating local lag trends, severely underestimating the magnitude and velocity of structural inflections.
- 2) Realistic Benchmarking for Temporal Causal Discovery: Cheng et al. [102] establish CausalTime (NeurIPS 2023), resolving a core bottleneck in temporal causal validation. Prior benchmarks relied on oversimplified linear autoregressive models with Gaussian noise, producing synthetic evaluations disconnected from real-world physics. CausalTime utilizes deep invertible generative neural networks and normalizing flows to capture complex nonlinear dynamics and heteroskedasticity while deriving exact ground-truth causal DAGs, creating a rigorous standard benchmark for evaluating causal graph discovery.
- 3) Principled Test-Time Adaptation Benchmarking (FAC): Wang et al. [101] critique existing TTA evaluations for subtle future-information leakage and ad-hoc batch scheduling. They establish a standardized leak-free streaming benchmark suite and introduce Frequency-Aware Calibration (FAC). FAC proves that adapting models directly in the raw temporal domain frequently amplifies high-frequency measurement noise into destructive gradient shocks; conversely, isolating adapter updates to orthogonal low- and mid-frequency Fourier sub-bands preserves base model stability while yielding consistent accuracy improvements across non-stationary streaming horizons.
- 4) Living Leaderboard Ecosystems (The TIME Benchmark): Addressing the rapid obsolescence of static benchmark datasets, Qiao et al. [90] introduce the TIME benchmark (50 fresh datasets, 98 tasks) and the public Hugging Face TIME-Leaderboard (<https://huggingface.co/spaces/Real-TSF/TIME-leaderboard>). By pairing automated weekly evaluation pipelines with pattern-level time

series decomposition, TIME evaluates foundation models across multi-scale periodicity, non-stationarity, and noise regimes, establishing a living, contamination-resistant community reference standard for zero-shot time-series intelligence.

7 Open Challenges and Future Horizons

Despite rapid progress in foundation models and LLM adaptations, temporal machine learning remains in a formative phase. In this section, we delineate six critical open research horizons that must be addressed to achieve truly universal temporal intelligence.

7.1 Irregular Sampling and Continuous-Time Dynamics

Existing foundation models (e.g., Chronos, TimesFM, MOIRAI) predominantly assume regularly sampled, synchronous discrete grids. However, real-world physical telemetry—such as ICU medical records, IoT sensor dropouts, and financial trade ticks—arrives at asynchronous, irregular time intervals. Emerging approaches such as FlowState [41] leverage continuous flow matching and continuous-time neural state-space models to achieve sampling-rate equivariance. Future foundation architectures must natively encode irregular time intervals (Δt) into latent manifolds rather than relying on crude imputation or resampling heuristics.

7.2 Prior-Data Fitted Bayesian Networks

Standard foundation models require thousands of gradient optimization steps during fine-tuning. A compelling alternative is demonstrated by TabPFN-TS [51], which adapts Prior-Data Fitted Networks to time series. By training a transformer exclusively on synthetic prior stochastic processes (e.g., Gaussian processes, Ornstein-Uhlenbeck, and Markov switching models), the network learns to perform Bayesian posterior inference in a single forward pass without parameter updates. Expanding synthetic prior spaces to encompass complex non-linear dynamics represents a fertile research direction.

7.3 Extreme Long-Horizon Context and Serial Scaling

In natural language, modern models handle context windows exceeding 1M tokens. In temporal analysis, historical series often span millions of high-frequency timestamps. Capturing decadal climate shifts, annual retail cycles, and second-level power grid vibrations within a single context window demands extreme context scaling. Architectural innovations such as long-context state-space models (Mamba), hierarchical linear attention, and serial scaling (as explored in Timer-XL [16] and Timer-S1 [18]) will be essential to eliminate context truncation.

7.4 Unified Multimodal Perception

Real-world decision-making rarely relies on isolated scalar numbers. An industrial operator monitors pressure gauges alongside diagnostic maintenance logs; a macroeconomist reads financial reports while analyzing stock tickers; an environmental scientist tracks satellite imagery across seasonal vegetation cycles [62]. Future foundation models must

TABLE 7
Systematic Synthesis of Test-Time Adaptation (TTA) and Causal Time-Series Foundation Models.

Framework	Key Reference	Core Mechanism & Adaptation / Causal Objective	Base Model	Param Footprint	Inference Overhead	Target Shift / Causal Capabilities
Category A: Test-Time Adaptation (TTA) under Non-Stationary Drift						
TSF-TTA	Kim et al. [66] (AAAI'25)	Online update of normalization statistics and linear projection heads via temporal consistency loss $\mathcal{L}_{TTA} = \ \hat{\mathbf{y}}_t - \mathcal{T}(\hat{\mathbf{y}}_{t-1})\ _2^2$	Frozen PatchTST / DLinear	Small (< 5M)	+12% latency	Abrupt mean/variance non-stationarity without source data
AdaNODEs	Dang et al. [67] (ICASSP'26)	Continuous latent trajectory adaptation via Neural ODEs: $\frac{d\mathbf{z}(t)}{dt} = f_\theta(\mathbf{z}(t), t)$ with self-supervised reconstruction	Any Frozen Forecaster	Minimal (0.15M)	+8% latency	Source-free continuous non-linear temporal drift
RG-TTA	Kumar et al. [68] (arXiv'26)	Regime-guided meta-control dynamically detecting stationary vs. non-stationary regimes, modulating gradient steps $\eta_t = \sigma(\text{RegimeGate}(X_t))$	Pretrained Transformer	Zero extra params	+5% latency	Multi-regime streaming series; prevents catastrophic forgetting
FAC Framework	Wang et al. [101] (arXiv'26)	Frequency-Aware Calibration projecting updates onto orthogonal spectral sub-bands to suppress high-frequency noise explosion	Pretrained TSFM	Lightweight	+10% latency	Leakage-free streaming adaptation; eliminates noisy gradient divergence
Category B: Causal Discovery, Structural Priors, and Counterfactual Foundation Models						
Causal-PT	Stein et al. [70] (arXiv'24)	Direct transformer mapping from multivariate temporal tokens to causal DAG adjacency matrix $\mathbf{A} \in \{0, 1\}^{D \times D}$ pre-trained across 10M SCMs	Transformer Encoder	25M	Single forward pass	Zero-shot discovery of directed causal graphs from observational series
CausalTimePrior	Thumm & Chen [71] (arXiv'26)	Prior-Data Fitted Network (PFN) pre-trained on 500M synthetic interventional TSCMs to approximate exact in-context posterior $P(\mathbf{Y} do(\mathbf{X}), \mathcal{D}_{obs})$	Transformer PFN	85M	Single forward pass	In-context causal effect estimation and zero-shot counterfactual forecasting
CaTSG	Xia et al. [72] (arXiv'25)	Structural causal score-based diffusion model conditioning reverse denoising trajectories on time-varying causal graphs $\mathcal{G}_t$	Causal Diffusion Net	45M	Multi-step ODE/SDE	Controllable trajectory generation under interventions ($do(X_t = \alpha)$)
CausalTime Bench	Cheng et al. [102] (NeurIPS'23)	Deep normalizing flow generator providing realistic multivariate time series paired with verifiable ground-truth causal graphs	Benchmark Generator	N/A	Offline evaluation	Ground-truth benchmark suite for time-series causal discovery algorithms
Jander Audit	Jander et al. [69] (arXiv'26)	Synthetic interventional audit harness probing persistence bias and structural distortion across foundation models	Chronos-2, TimesFM, MOIRAI	N/A	Offline audit	Uncovers persistence bias and regime switch degradation in TSFMs

advance beyond ad-hoc text reprogramming to train native multi-modal architectures from scratch—treating numerical time series, textual reports, spatial optical imagery, and spectral vibrations as co-equal perceptual modalities, as pioneered by Chronicle [87], VLT [86], and ChronoSteer [88].

7.5 Calibrated Uncertainty under Distribution Shift

In safety-critical applications (such as healthcare and grid management), an uncalibrated point forecast is hazardous. As demonstrated by recent reliability audits [26], current zero-shot models frequently underestimate predictive variance when confronted with unprecedented macroeconomic shocks or extreme weather events. Developing conformal prediction techniques, robust Bayesian heads, and flow-based generative sampling that maintain rigorous coverage guarantees under out-of-distribution shifts is imperative.

7.6 Edge Deployment, Hardware Quantization, and Green AI

While research models scale to billions of parameters, industrial IoT environments require deployment on low-power microcontrollers, edge gateways, and automotive Electronic Control Units (ECUs). Systematic cross-platform benchmarking (HoliBench [98]), integer mixed-precision quantization (INT8/INT4 on embedded FPGAs [99]), ultra-compact patch encoders [63], [65], and carbon-aware battery-buffered dynamic dispatch (FM-CAC [100]) will be essential to trans-

form temporal foundation research into sustainable, energy-efficient cyber-physical systems.

7.7 Autonomous Living Foundations and Continuous Discovery

With dozens of new preprints posted weekly across machine learning repositories, academic surveys quickly become obsolete. Future research must embrace autonomous, living benchmark harnesses (e.g., LiveHouse-TS [95]) and continuous automated snowballing pipelines that monitor arXiv and Semantic Scholar streams, evaluate newly dropped weights against standardized multi-granularity leakage audits (It's TIME [90]), and iteratively update systematic taxonomies.

8 Conclusion

In this survey, conducted under the PRISMA 2020 systematic review protocol, we have synthesized the foundational developments at the intersection of Large Language Models and Time Series Foundation Models from 2021 through 2026. We formalized mathematical definitions of temporal sequences, subseries patching, and quantization; introduced a unified five-dimensional taxonomy; and compared 41 core benchmark and foundation models across architecture, tokenization, pre-training corpora, and uncertainty formulations.

Our critical synthesis highlights a decisive architectural demarcation:

- Native TSFMs (such as Chronos, TimesFM, MOIRAI, Timer, and TTM) demonstrate superior parameter efficiency, faster inference latency, and stronger zero-shot numeric forecasting accuracy by training directly on temporal dynamics without linguistic mismatch.
- Repurposed LLMs (such as Time-LLM and multimodal reasoning frameworks) excel not at raw numeric extrapolation, but as cognitive orchestrators—grounding numerical time series within textual narratives, metadata, and decision-support systems.

Looking ahead, the frontier of temporal intelligence lies in resolving benchmark contamination, achieving continuous-time sampling equivariance, calibrating uncertainty under non-stationary shifts, and delivering ultra-efficient edge deployment. We hope this survey provides a rigorous compass for researchers and practitioners building the next generation of universal temporal models.

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